

THE COMBINATORICS OF SECTOR RENORMALIZATION

WILLIE RUSH LIM

ABSTRACT. The goal of this note is to systematically develop the fundamental arithmetic and combinatorial properties of the sector renormalization operation on rigid rotations. We identify the specific framework of modified continued fractions appropriate for sector renormalization and analyze their properties. By allowing infinite first return times, this framework yields a dynamical compactification characterized by a universal property. We also discuss the corresponding natural extension and introduce the notion of a time (semi-)group. For example, we demonstrate how a bi-infinite tower of sector renormalizations of irrational rotations can be packaged within a single dynamical plane as a cascade of translations. This note will serve as a foundational combinatorial tool for studying the geometric properties of sector renormalizations of holomorphic maps with irrationally indifferent fixed points, particularly neutral quadratic polynomials.

CONTENTS

1. Introduction	1
2. Renormalization of irrational rotations	7
3. Diophantine approximation	12
4. The compactification	17
5. The natural extension	26
References	32

1. INTRODUCTION

Let

$$\Theta := \left(-\frac{1}{2}, \frac{1}{2} \right) \setminus \mathbb{Q}.$$

While the dynamics of an irrational rotation

$$\mathbb{r}_\theta : z \mapsto e^{2\pi i\theta} z$$

are traditionally classified via the entries a_n of its regular continued fraction expansion, this note explores an alternative symbolic coding arising from *sector renormalization*.

Geometrically, let $S \subset \overline{\mathbb{D}}$ be a fundamental sector of $\mathbb{r}_\theta$ inside the closed unit disk that is cut out by a pair of radial arcs subtending an angle of $2\pi|\theta|$ radians. The power map $\psi(z) = z^{1/|\theta|}$ glues the radial sides of S together, mapping the sector back onto $\overline{\mathbb{D}}$. The first return map of $\mathbb{r}_\theta$ to S is a piecewise continuous map

consisting of a pair of iterates $(r_{\theta}^{\bar{a}}, r_{\theta}^{\bar{a}+1})$. Under ψ , this pair projects onto a new rotation $r_{\mathbf{g}(\theta)} : \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$, where the map $\mathbf{g} : \Theta \rightarrow \Theta$ is uniquely defined by the relation

$$\mathbf{g}(\theta) \equiv -\frac{1}{\theta} \pmod{1}.$$

Iterating this operation yields a symbolic representation of the irrational θ by an element $\langle(\varepsilon_n, \bar{a}_n)\rangle_{n \geq 1}$ in the sequence space $(\{-1, +1\} \times \mathbb{N}_{\geq 2})^{\mathbb{N}}$, where ε_{n+1} and $\bar{a}_{n+1}$ track the orientation and first return time at renormalization level n .

This construction provides a combinatorial model for the classical sector renormalization of arbitrary holomorphic germs pioneered by Douady–Ghys [Dou87] and Yoccoz [Yoc95]. It connects to the modified continued fractions commonly utilized in near-parabolic renormalization theory; see Section 1.3 for historical remarks.

While these arithmetic approaches are often considered “routinely” equivalent, we emphasize the importance of distinguishing 0^+ and 0^- . We discuss the appropriateness of the compactification $\Theta \rightsquigarrow \overline{\Theta}$ by allowing entries to take the value ∞ and we justify its merit in Theorem B by establishing two universal properties of $\overline{\Theta}$. We also analyze the corresponding inverse limit $\varprojlim \overline{\Theta}$ for applications to the unified theory of neutral renormalization. Ultimately, our goal is to systematically describe:

- (I) the dynamical and arithmetic aspects of $\mathbf{g} : \Theta \rightarrow \Theta$;
- (II) the appropriate dynamical compactification of $(\Theta, \mathbf{g})$;
- (III) the properties of the natural extension of $\mathbf{g}$.

1.1. Modified continued fractions. The arithmetic properties of $\mathbf{g}$ boil down to a variant of the nearest integer continued fraction expansion. It first appeared in the work of Hurwitz [Hur89] in 1889. We take a more dynamical point of view and emphasize that the expansion is parameterized by the infinite sequence space

$$\Sigma^{\mathbb{N}} := \left\{ \langle(\varepsilon_n, \bar{a}_n)\rangle_{n \geq 1} : \varepsilon_n \in \{-1, +1\}, \bar{a}_n \in \mathbb{N}_{\geq 2} \right\}.$$

For any sequence of non-zero integers $x_1, x_2, x_3, \dots$, denote

$$(1.1) \quad [x_1, x_2, x_3, \dots]_- := \frac{1}{x_1 - \frac{1}{x_2 - \frac{1}{x_3 - \dots}}}.$$

When $|x_n| \geq 2$ for cofinitely many $n \geq 1$, the expression above is a well-defined irrational number. One of the theorems in this note is the following.

Theorem A. *The function $\mathfrak{X} : \Sigma^{\mathbb{N}} \rightarrow \Theta$ given by*

$$(1.2) \quad \mathfrak{X}(\langle(\varepsilon_n, \bar{a}_n)\rangle_{n \geq 1}) = [\varepsilon_1 b_1, \varepsilon_2 b_2, \varepsilon_3 b_3, \dots]_- \quad \text{where} \quad b_n = \bar{a}_n + \frac{1 + \varepsilon_n \varepsilon_{n+1}}{2}$$

is a homeomorphism conjugating the standard shift map $\mathfrak{s} : \Sigma^{\mathbb{N}} \rightarrow \Sigma^{\mathbb{N}}$ with the map $\mathbf{g} : \Theta \rightarrow \Theta$.

The proof relies on the study of Diophantine approximations related to the modified continued fraction above. For $\sigma = \langle(\varepsilon_n, \bar{a}_n)\rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$ and b_n ’s being as introduced in the theorem, we can write

$$[\varepsilon_1 b_1, \varepsilon_2 b_2, \dots, \varepsilon_n b_n]_- = \frac{p_{[n]}}{q_{[n]}},$$

where $p_{[n]} = p_{[n]}(\sigma)$ and $q_{[n]} = q_{[n]}(\sigma)$ are co-prime integers and $q_{[n]} \geq 1$. The $q_{[n]}$ ’s can be characterized in many other ways including the first return times of

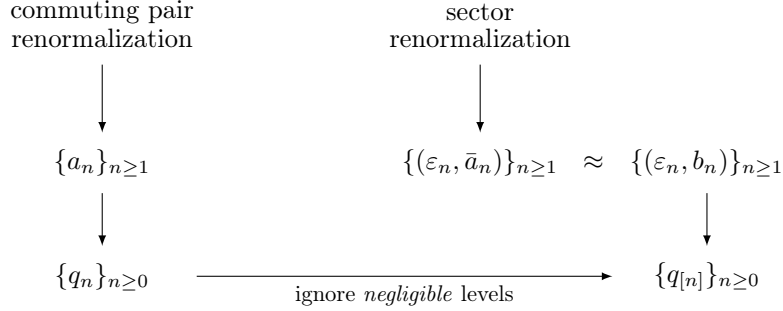


FIGURE 1. Relationship between the different codings

an iterated pre-renormalization (Proposition 2.3). Theorem A follows from the arithmetic properties of these convergents.

In Section 3, we contrast our approach with the regular continued fraction expansion (the standard a_n 's) and the nearest integer continued fraction expansion. We establish a conversion to the regular one, which will come in handy in subsequent applications. The regular continued fraction expansion was related to commuting pair renormalization in the theory of circle maps, whereas the nearest integer variant is similar to our expansion and has also been used several times in complex dynamics; see Section 1.3 for a historical account. Figure 1 summarizes the relationships between them. Note that even though the b_n 's are more desirable in the number-theoretic point of view, they satisfy an intricate admissibility rule (depending on $\varepsilon_n \varepsilon_{n+1}$) and are not easily describable from the geometric procedure of sector renormalization.

1.2. Compactification and natural extension. Below are three embeddings of the space of irrationals Θ into compact spaces that are of particular interest.

- The modular embedding $\mu : \Theta \rightarrow \mathbb{T} = \mathbb{R}/\mathbb{Z}$.
This is the naive embedding given by $\mu(\theta) = \theta \pmod{1}$.
- The segment embedding $\eta : \Theta \rightarrow [0, 1]$.
This is given by $\eta(\theta) = \theta$ if $\theta > 0$ and $\eta(\theta) = \theta + 1$ if $\theta < 0$. It satisfies the following commutative diagram.

$$\begin{array}{ccc}
\Theta & \xrightarrow{\eta} & [0, 1] \\
& \searrow \mu & \downarrow 0 \sim 1 \\
& & \mathbb{T}
\end{array}$$

The map $\mathbf{g}$ naturally extends to a self-map of $\mathbb{T}$ that is continuous everywhere except at the origin, at which wild oscillations are produced on both the left and the right side. See Figure 3. Unlike μ , the embedding η has the advantage of distinguishing 0^+ and 0^- .

- The dynamical embedding $\text{ord} : \Theta \rightarrow \Omega$.
As the name suggests, this map has direct ties to the applications in neutral renormalization. (See Section 1.3.) Geometrically, it encodes the cyclic order of forward orbits of successive pre-renormalizations of $\mathbf{r}_\theta$. Here is how it works. Given any two positive integers k and l , let $\Omega_{k,l}$ denote the set of maps $f : \{0, 1, \dots, k\}^{l+1} \rightarrow \mathbb{T}$ modulo isotopy of the image of f in $\mathbb{T}$.

This set corresponds to the set of all circular permutations of s elements where $1 \leq s \leq ((k+1)^{l+1} - 1)!$. The set $\Omega_{k,l}$ is finite and it will be equipped with the discrete topology. Consider the infinite product space

$$\Omega = \prod_{k,l \geq 1} \Omega_{k,l}.$$

Given an irrational $\theta \in \Theta$, we denote by $\text{ord}_{k,l}(\theta) \in \Omega_{k,l}$ the ambient isotopy class of the embedding

$$(j_0, \dots, j_l) \mapsto \mathbb{r}_\theta^{j_0 q_{[0]}(\theta) + \dots + j_l q_{[l]}(\theta)}(1).$$

The maps $\text{ord}_{k,l} : \Theta \rightarrow \Omega_{k,l}$ altogether produce the embedding

$$\text{ord} : \Theta \rightarrow \Omega, \quad \text{ord}(\theta) = \{\text{ord}_{k,l}(\theta)\}_{k,l}.$$

Via the map $\mathfrak{X}$ from Theorem A, the shift map $\mathfrak{s}$ on the compact sequence space

$$\overline{\Theta} := \left\{ \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} : \varepsilon_n \in \{-1, +1\}, \bar{a}_n \in \mathbb{N}_{\geq 2} \cup \{\infty\} \right\}$$

is a compactification of $(\Theta, \mathfrak{g})$. The three embeddings listed above indeed extend to continuous maps

$$\bar{\mu} : \overline{\Theta} \rightarrow \mathbb{T}, \quad \bar{\eta} : \overline{\Theta} \rightarrow [0, 1], \quad \overline{\text{ord}} : \overline{\Theta} \rightarrow \Omega$$

respectively. We prove:

Theorem B (Universal property of $\overline{\Theta}$). *The compactification $(\overline{\Theta}, \mathfrak{s})$ of $(\Theta, \mathfrak{g})$ is uniquely characterized by one of the following properties.*

- (1) $(\overline{\Theta}, \mathfrak{s})$ is the smallest compactification of $(\Theta, \mathfrak{g})$ that respects η , and it has the universal property that every other compactification of $(\Theta, \mathfrak{g})$ respecting η factors to $(\overline{\Theta}, \mathfrak{s})$.
- (2) $(\overline{\Theta}, \mathfrak{s})$ is the smallest compactification of $(\Theta, \mathfrak{g})$ that respects ord , and it has the universal property that every other compactification of $(\Theta, \mathfrak{g})$ respecting ord factors to $(\overline{\Theta}, \mathfrak{s})$.

Consequently, every compactification of $(\Theta, \mathfrak{g})$ that respects η also respects ord , and vice versa.

A more precise formulation can be found in Section 4.2. In spirit, it is the dynamical opposite of Stone-Ćech compactification in topology. The minimality property fails for the modular embedding (Proposition 4.10).

For each element θ of $\overline{\Theta}$, we can assign a totally ordered additive abelian group called the *time group* $(\mathcal{T}_\theta^{\text{gp}}, >)$ of θ whose generators are a generalization of the denominator continuants $q_{[n]}$'s discussed above. The positive cone $\mathcal{T}_\theta = \{p \in \mathcal{T}_\theta^{\text{gp}} : p > 0\}$, which we call the *time semigroup*, will be important in applications.

In Section 5, we study the natural extension $\mathfrak{s} : \overline{\Theta} \rightarrow \overline{\Theta}$, which is the bi-infinite shift space

$$\mathfrak{s} : \overline{\Theta} \rightarrow \overline{\Theta}$$

obtained by taking the inverse limit. We show that each element $\underline{\theta}$ of $\overline{\Theta}$ induces a totally ordered group $(\mathbf{T}_{\underline{\theta}}^{\text{gp}}, >)$ with a countably infinite number of generators $Q_{[n]}$, $n \in \mathbb{Z}$ mimicking the properties of denominator continuants. Again, the *time semigroup* $\mathbf{T}_{\underline{\theta}} = \{P \in \mathbf{T}_{\underline{\theta}}^{\text{gp}} : P > 0\}$ is useful in applications.

Throwing out ∞ gives us the shift-invariant subspace $\underline{\Theta}$ of $\overline{\Theta}$. Every element of $\underline{\Theta}$ induces a unique bi-infinite tower of sector renormalizations of irrational rotations. In Section 5.2, we describe how this tower can be fit into a single dynamical plane as a group of real translations, called *cascades*. Translational cascades with periodic combinatorics first appeared in [DL23, Section 2].

1.3. Motivation and historical remarks. This note provides the background on the combinatorics of neutral renormalization in holomorphic dynamics.

1.3.1. On sector renormalization. As previously mentioned, sector renormalization for neutral holomorphic germs was initially studied by Douady-Ghys, and Yoccoz. Variants of sector renormalization for neutral quadratic polynomials $f_\theta(z) = e^{2\pi i\theta}z + z^2$ have been studied under some arithmetic conditions. For bounded type irrationals θ , Dudko, Lyubich, and Selinger [DLS20, DL23] studied Siegel-Pacman renormalization to prove parameter universality and local connectivity of the Mandelbrot set near Siegel parameters. For high type irrationals θ , Inou and Shishikura studied near-parabolic renormalization [IS06] which resulted in a number of significant applications, e.g. [BC12, CC15, Che13, Che19, SY24, Che25]. A variant of the modified continued fraction expansion presented in Theorem A has appeared a number of times in holomorphic dynamics, e.g. [Yoc95, IS06, CC15, AC17, Che25]; see Section 3.2.

In [DL26], Dudko and Lyubich showed that sector renormalizations of neutral quadratic polynomials $f_\theta(z) = e^{2\pi i\theta}z + z^2$ with irrational rotation numbers θ can be constructed in such a way that the resulting maps $\mathcal{R}_{\text{sec}}^n f_\theta$ are pre-compact. In particular, their critical orbit is well-contained in its domain of definition. (Their result is based on some result of uniform *a priori bounds* [DL22] which was originally formulated using the regular a_n 's.) Every element in the closure of the space of renormalization orbits $\{\langle \mathcal{R}_{\text{sec}}^{n+m} f_\theta \rangle_{n \geq 0} : m \geq 0, \theta \in \Theta\}$ has combinatorics that can be uniquely encoded by an element $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}$ of the compactification $\overline{\Theta}$, and it also comes with an associated full Lavaurs-Épstein renormalization tower that can be parametrized by the time semigroup $\mathcal{T}_\theta$ of θ . Any term $\bar{a}_{n+1} = \infty$ indicates the occurrence of simple parabolic bifurcation at renormalization level n .

The extension $\bar{\eta}$ reflects the role of orientation in distinguishing top versus bottom Lavaurs parabolic enrichments, while the fiber structure of $\bar{\mu}$ encapsulates the need for parabolic blow-ups of the Mandelbrot set to define a geometric topology where Julia sets are continuous. Meanwhile, the extension ord ensures that the cyclic arrangement of critical orbits under successive pre-renormalizations varies continuously. Together, these properties provide the framework necessary for further applications, including the uniform continuity of Mother Hedgehogs [DL26], the zero-area theorem for postcritical sets [Lim26b], and the rigidity of the attractor of renormalization of neutral quadratic polynomials [DLL26].

Let us elaborate on the last point above. Elements of the attractor are bi-infinite towers $\langle f_n \rangle_{n \in \mathbb{Z}}$ of sector renormalization and their combinatorial data are elements of the natural extension $\underline{\Theta}$. With Dzmitry Dudko and Mikhail Lyubich, we prove:

Theorem 1.1 ([DLL26]). *The full renormalization attractor for neutral quadratic polynomials is combinatorially rigid. In particular, modulo conformal conjugacy, the renormalization attractor is conjugate to the shift map $\mathfrak{s} : \underline{\Theta} \rightarrow \underline{\Theta}$.*

In the proof, we bring a bi-infinite tower of renormalizations $\langle f_n \rangle_{n \in \mathbb{Z}}$ to a single dynamical plane of a semigroup $\mathbf{F}$ of σ -proper holomorphic maps onto $\mathbb{C}$ called a *neutral cascade* which can be parametrized by the time semigroup $\mathbf{T}_\theta$ of θ . (For periodic combinatorics, this cascade was studied in [McM98, DL23].) We then prove uniform complex bounds for neutral cascades and apply them to construct an affine conjugacy between any two combinatorially equivalent neutral cascades. Neutral cascades are non-linear analogues of translational cascades discussed in Section 5.2. Under some arithmetic condition (called the Brjuno condition), $\mathbf{F}$ admits an invariant topological half-plane (called the Siegel set) in which $\mathbf{F}$ is analytically conjugate to a translational cascade (c.f. Section 5.2).

1.3.2. *On commuting pair renormalization.* Lastly, let us comment on the connection between the renormalization theory of neutral quadratic polynomials and that of real-analytic (uni-)critical circle maps with irrational rotation number.

In the theory of critical circle maps, renormalization is defined in terms of commuting pairs supported on real intervals and the combinatorics can be described in terms of the regular continued fraction expansion $\{a_n\}_{n \geq 1}$ together with the classical Gauss map $G(\theta) = \{\frac{1}{\theta}\}$. It was initially observed by Lanford [Lan88] that the study of commuting pair renormalization would explain the rigidity and universality phenomenon for critical circle maps. This has now been justified by de Faria, de Melo, and Yampolsky [dF99, dFdM99, Yam99, Yam01, Yam02, Yam03].

The analog of Theorem 1.1 was proven by Yampolsky [Yam01] where the corresponding renormalization horseshoe was shown to be conjugate to the shift map on $\{1, 2, 3, \dots, \infty\}^{\mathbb{Z}}$. The term ∞ again encodes the degeneration coming from simple parabolic bifurcation, but, unlike sector renormalization, real-symmetry prevents any possible ambiguity in orientation. Every element of the attractor also admits the structure of a cascade of real-symmetric σ -proper maps onto $\mathbb{C}$. (For periodic combinatorics, this cascade was studied in [Lim26a].) In [Yam02, Yam03], the commuting-pair renormalization was upgraded to cylinder renormalization, a gluing operation that is geometrically similar to sector renormalization. Within this new framework of cylinder renormalization, Yampolsky [Yam02, Yam03] successfully brought Lanford's Program to completion.

There is a surgery operation, invented by Douady and Ghys [Ghy84, Dou87], that converts a particular cubic rational critical circle map to a neutral quadratic polynomial f_θ for any bounded type θ . This connection to circle maps is responsible for many significant results on the dynamics of f_θ in the bounded type regime [Pet96, McM98, DL22] and, together with its convenience, justifies the wide use of the regular a_n 's in the field of neutral dynamics as a whole.

1.4. **Acknowledgments.** I would like to thank Dzmitry Dudko for highlighting the importance of ord. A significant part of this note was written during my visit to the Banach Center during the Simons Semester on Continued Fractions, Fractals, Ergodic theory and Dynamics in May 2026. This work was partially supported by the Simons Foundation grant (award no. SFI-MPS-T-Institutes-00010825) and from State Treasury funds as part of a task commissioned by the Minister of Science and Higher Education under the project "Organization of the Simons Semesters at the Banach Center – New Energies in 2026–2028" (agreement no. MNiSW/2025/DAP/491).

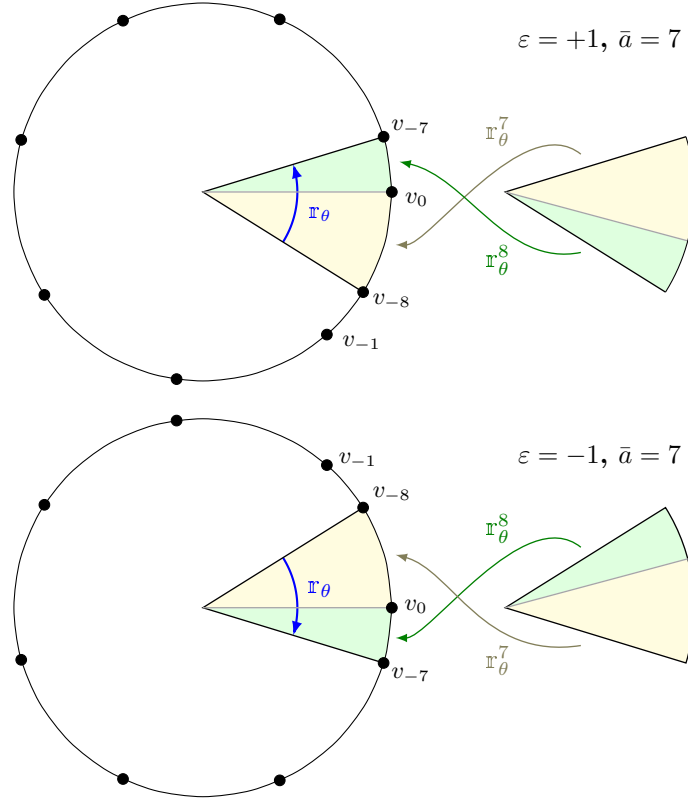


FIGURE 2. Two examples of the first return map on the shaded sector S_θ .

2. RENORMALIZATION OF IRRATIONAL ROTATIONS

2.1. Sector renormalization. For any irrational $\theta \in \Theta$, denote

$$\varepsilon(\theta) := \text{sgn}(\theta) \in \{-1, +1\} \quad \text{and} \quad \bar{a}(\theta) := \left\lfloor \frac{1}{|\theta|} \right\rfloor \in \mathbb{N}_{\geq 2}.$$

The sign $\varepsilon = \varepsilon(\theta)$ tells us whether or not r_θ rotates clockwise or anticlockwise. The number $\bar{a} = \bar{a}(\theta)$ is the unique positive integer such that

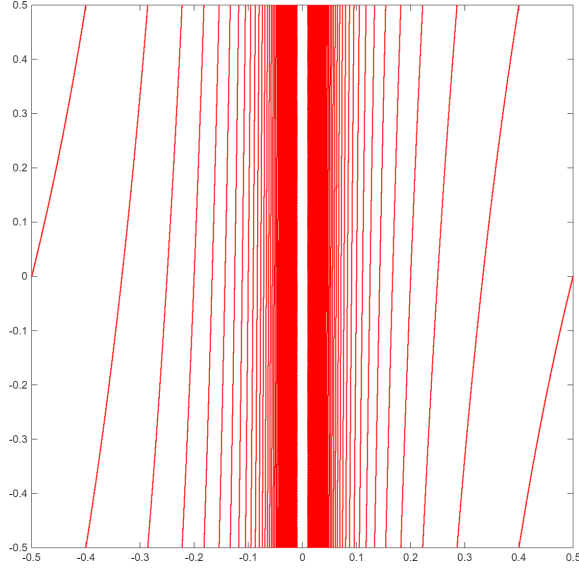
$$\bar{a}|\theta| < 1 < (\bar{a} + 1)|\theta|.$$

Let us first fix $\theta \in \Theta$. For $k \in \mathbb{Z}$, denote

$$v_k = v_k(\theta) := r_\theta^k(1).$$

For any two distinct integers $k, l \in \mathbb{Z}$, we denote by $\triangle_\theta(k, l)$ the closed radial sector bounded by the radial line segment from 0 to v_k , the radial line segment from 0 to v_l , and the shortest circular arc in $\partial\mathbb{D}$ joining v_k and v_l . We will in particular consider the sector

$$S_\theta := \triangle_\theta(-\bar{a} - 1, -\bar{a}).$$

FIGURE 3. The graph of $\mathbf{g}$

It contains the point $v_0 = 1$. The first return map of $\mathbf{r}_\theta$ back to S_θ is the piecewise linear map

$$\text{FRM}_\theta(z) = \begin{cases} \mathbf{r}_\theta^{\bar{a}+1}(z) & \text{if } z \in \triangle_\theta(-\bar{a}-1, -2\bar{a}-1), \\ \mathbf{r}_\theta^{\bar{a}}(z) & \text{if } z \in \triangle_\theta(-2\bar{a}-1, -\bar{a}). \end{cases}$$

See Figure 2 for an illustration. The power map

$$\psi_\theta : S_\theta \rightarrow \overline{\mathbb{D}}, \quad \psi_\theta(z) = z^{1/|\theta|}$$

sends the interior of S_θ conformally onto the unit disk minus the radial slit $\gamma_\theta = \{\arg z = |\theta|^{-1} \arg v_{-\bar{a}}\}$, and it glues the two radial edges of S_θ together onto the slit. Under ψ_θ , FRM_θ projects to a new rigid rotation $\mathbf{r}_{\mathbf{g}(\theta)} : \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$ for some irrational $\mathbf{g}(\theta) \in \Theta$. The slit γ_θ is actually equal to $\{\arg z = -2\pi \mathbf{g}(\theta)\}$. The map

$$\mathbf{g} : \Theta \rightarrow \Theta$$

should be thought as a modified Gauss map. We call $\mathbf{r}_{\mathbf{g}(\theta)}$ the *sector renormalization* of $\mathbf{r}_\theta$.

Lemma 2.1. *For $\theta \in \Theta$, $\mathbf{g}(\theta)$ is the unique irrational in Θ such that*

$$\mathbf{g}(\theta) \equiv -\frac{1}{\theta} \pmod{1}.$$

More precisely, we have

$$\mathbf{g}(\theta) = -\frac{1}{\theta} + \varepsilon(\theta) b(\theta) \quad \text{where } b(\theta) = \bar{a}(\theta) + \frac{1 + \varepsilon(\theta) \varepsilon(\mathbf{g}(\theta))}{2}.$$

The map $\mathbf{g} : \Theta \rightarrow \Theta$ is an infinite-to-one surjective continuous function.

The graph of $\mathbf{g}$ is illustrated in Figure 3.

Proof. Denote $\varepsilon_1 = \varepsilon(\theta)$, $\varepsilon_2 = \varepsilon(\mathbf{g}(\theta))$, and $\bar{a} = \bar{a}(\theta)$. The first return map back to S_θ can be written as the pair of maps $\mathbf{r}_{\bar{a}\theta - \varepsilon_1}$ and $\mathbf{r}_{(\bar{a}+1)\theta - \varepsilon_1}$ where both $\bar{a}\theta - \varepsilon_1$ and $(\bar{a}+1)\theta - \varepsilon_1$ are contained in $(-\frac{1}{2}, \frac{1}{2})$. Therefore, for $z \in \overline{\mathbb{D}}$, there is some $k = k(z) \in \{\bar{a}, \bar{a}+1\}$ such that

$$\mathbf{r}_{\mathbf{g}(\theta)}(z) = \psi_\theta \circ \mathbf{r}_{k\theta - \varepsilon_1} \circ \psi_\theta^{-1}(z) = e^{2\pi i(k\theta - \varepsilon_1)/|\theta|} z = e^{2\pi i(\varepsilon_1 k - \frac{1}{\theta})} z,$$

so then $\mathbf{g}(\theta) \equiv -\frac{1}{\theta} \pmod{1}$.

Let us write $\mathbf{g}(\theta) = -\frac{1}{\theta} + x$ where $x = x(\theta)$ is the unique integer closest to $\frac{1}{\theta}$. Since $\bar{a} < \frac{\varepsilon_1}{\theta} < \bar{a}+1$, then either $x = \varepsilon_1 \bar{a}$ or $x = \varepsilon_1(\bar{a}+1)$. If $x = \varepsilon_1 \bar{a}$, then

$$\varepsilon_1 \mathbf{g}(\theta) = -\frac{\varepsilon_1}{\theta} + \varepsilon_1 x = -\frac{\varepsilon_1}{\theta} + \bar{a} < 0.$$

which implies that $\varepsilon_1 \varepsilon_2 = -1$. Else, if $x = \varepsilon_1(\bar{a}+1)$, then

$$\varepsilon_1 \mathbf{g}(\theta) = -\frac{\varepsilon_1}{\theta} + \varepsilon_1 x = -\frac{\varepsilon_1}{\theta} + \bar{a} + 1 > 0$$

and so $\varepsilon_1 \varepsilon_2 = +1$. In both cases, we do have $x = \varepsilon_1 b(\theta)$.

The function $\mathbf{g}$ is continuous since $-\frac{1}{\theta}$ and $x(\theta)$ are continuous in $\theta \in \Theta$. It is clear that $\mathbf{g}$ is infinite-to-one and surjective. $\square$

2.2. More notation. Denote $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\Sigma := \{-1, +1\} \times \mathbb{N}_{\geq 2}$. Consider the infinite sequence space

$$\Sigma^{\mathbb{N}} = \{ \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} : (\varepsilon_n, \bar{a}_n) \in \Sigma \text{ for all } n \geq 1 \},$$

equipped with the infinite product topology. For every $\sigma = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$, we denote

$$b_n = b_n(\sigma) := \bar{a}_n + \frac{1 + \varepsilon_n \varepsilon_{n+1}}{2} \quad \text{for } n \geq 1.$$

Associated to σ are the pair of sequences $\{p_{[n]} = p_{[n]}(\sigma)\}_{n \geq 0}$ and $\{q_{[n]} = q_{[n]}(\sigma)\}_{n \geq 0}$ where

$$p_{[0]} = 0, \quad q_{[0]} = 1, \quad p_{[1]} = \varepsilon_1, \quad q_{[1]} = b_1,$$

and recursively for $n \geq 2$,

$$(2.1) \quad p_{[n]} = b_n p_{[n-1]} - \varepsilon_{n-1} \varepsilon_n p_{[n-2]} \quad \text{and} \quad q_{[n]} = b_n q_{[n-1]} - \varepsilon_{n-1} \varepsilon_n q_{[n-2]}.$$

We have a well-defined function

$$\mathfrak{Y} : \Theta \rightarrow \Sigma^{\mathbb{N}}, \quad \theta \mapsto \langle (\varepsilon(\mathbf{g}^{n-1}(\theta)), \bar{a}(\mathbf{g}^{n-1}(\theta))) \rangle_{n \geq 1}.$$

Throughout the rest of this section, we will fix an irrational number $\theta \in \Theta$. We denote

$$\langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} = \mathfrak{Y}(\theta) \quad \text{and} \quad b_n = b_n(\mathfrak{Y}(\theta)) \quad \text{for all } n \geq 1.$$

For $n \geq 0$, we will denote

$$p_{[n]} \equiv p_{[n]}(\theta) = p_{[n]}(\mathfrak{Y}(\theta)), \quad q_{[n]} \equiv q_{[n]}(\theta) = q_{[n]}(\mathfrak{Y}(\theta)), \quad \mathbf{r}_\theta^{[n]} \equiv \mathbf{r}_\theta^{q_{[n]}},$$

and

$$\theta_n = \mathbf{g}^n(\theta).$$

By Lemma 2.1, we have

$$(2.2) \quad \theta_n = -\frac{1}{\theta_{n-1}} + \varepsilon_n b_n \quad \text{for all } n \geq 1.$$

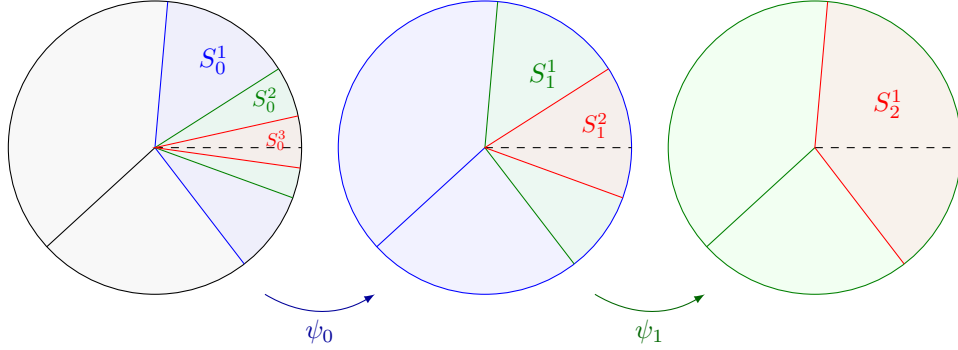


FIGURE 4. A nest of renormalization sectors

2.3. Iterated sector renormalizations. For any $m \geq 0$ and $n \in \mathbb{Z}$, denote

$$S_m^1 = S_{\theta_m}, \quad \psi_{\theta_m} = \psi_m, \quad \text{and} \quad I_m^n := \{z \in \overline{\mathbb{D}} : \arg z = \arg r_{\theta_m}^n(1)\}.$$

Observe that the radial slit γ_{θ_m} discussed previously is equal to I_{m+1}^{-1} and is always disjoint from the sector $S_{m+1,1}$. Hence, the sector S_{m+1}^1 can be lifted under the gluing map ψ_m to a sector S_m^2 contained in S_m^1 . By repeating this process, we obtain the nest of sectors

$$S_m^1 \supset S_m^2 \supset S_m^3 \supset S_m^4 \supset \dots$$

where

$$S_m^n := \psi_{\theta_m}^{-1}(S_{m+1}^{n-1}) \quad \text{for all } m \geq 0 \text{ and } n \geq 2.$$

For every $n \geq 1$, the angular width of S_0^n is equal to

$$l_{[n-1]} := |\theta_0 \theta_1 \dots \theta_{n-1}|.$$

For convenience, we will also set $l_{[-1]} = 1$.

Proposition 2.2. *For all $n \geq 1$,*

- (1_n) $l_{[n-2]} = b_n l_{[n-1]} - \varepsilon_n \varepsilon_{n+1} l_{[n]},$
- (2_n) $1 = q_{[n]} l_{[n-1]} - \varepsilon_n \varepsilon_{n+1} q_{[n-1]} l_{[n]},$
- (3_n) $\varepsilon_{n+1} l_{[n]} = q_{[n]} \theta - p_{[n]}.$

Proof. We can rewrite (2.2) as

$$(2.3) \quad \varepsilon_n = b_n \theta_{n-1} - \varepsilon_{n+1} |\theta_{n-1} \theta_n|.$$

Multiplying this equation by $\varepsilon_n l_{[n-2]}$ gives us (1_n). Observe that when $n = 1$, the equation above gives us (2₁) and (3₁) as well. For $n \geq 2$, (2_n) follows directly from (2_{n-1}) and the recurrence relation (2.1). Observe also that (3₀) holds when the left hand side is to be taken to be θ . For $n \geq 2$, (3_n) follows from (1_n), (3_{n-1}), (3_{n-2}), and the recurrence relations (2.1) as follows:

$$\begin{aligned} \varepsilon_{n+1} l_{[n]} &= b_n \varepsilon_n l_{[n-1]} - \varepsilon_n l_{[n-2]} \\ &= b_n (q_{[n-1]} \theta - p_{[n-1]}) - \varepsilon_{n-1} \varepsilon_n (q_{[n-2]} \theta - p_{[n-2]}) \\ &= q_{[n]} \theta - p_{[n]}. \end{aligned} \quad \square$$

These $q_{[n]}$'s are known as the first return times of θ due to the following proposition.

Proposition 2.3 (Dynamical characterization of $q_{[n]}$'s). *For all $n \geq 1$,*

- (1) $r_\theta^{[n]}$ *is the rigid rotation by angle $\varepsilon_{n+1}l_{[n]} = |\theta_0 \dots \theta_{n-1}|\theta_n$;*
- (2) *the sector S_0^n is bounded by radial segments $I_0^{-q_{[n]} + \varepsilon_n \varepsilon_{n+1} q_{[n-1]}}$ and $I_0^{-q_{[n]}}$;*
- (3) $q_{[n]}$ *is the unique smallest positive integer that satisfies*

$$|[1, r_\theta^{[n]}(1)]| \leq \frac{1}{2} |[1, r_\theta^{[n-1]}(1)]|$$

where $|\cdot|$ refers to the normalized angular measure on the unit circle.

In neutral renormalization theory in holomorphic dynamics, the inequality in (3) gives uniform *improvement of domain*.

We will take a dynamical approach to prove the proposition. We will make use of a nest of dynamical triangulations $\{\Delta_m^n\}_{n \geq 0}$ of the closed unit disk for all $m \geq 0$. Each Δ_m^n is a collection of closed sectors in $\mathbb{D}$ with vertex at 0 such that their union is $\mathbb{D}$ and that they have pairwise disjoint interiors. By a nest, we mean that for every n , every sector in Δ_m^n is a union of finitely many sectors in Δ_m^{n+1} . Here is how these triangulations are constructed.

The level 0 triangulation Δ_m^0 is defined by the union of the two closed sectors formed by cutting $\mathbb{D}$ along I_m^0 and I_m^{-1} . We will denote by $A_{m,0}$ the sector in Δ_m^0 with angular width $1 - |\theta_m|$ and by $A_{m,1}$ the sector in Δ_m^0 with angular width $|\theta_m|$.

For all $m \geq 0$ and $n \geq 1$, the sector S_m^n is the union of two triangles $A_{m,0}^n$ and $A_{m,1}^n$ which are the lifts of $A_{m+n,0}$ and $A_{m+n,1}$ under the map $\psi_{m+n-1} \circ \dots \circ \psi_m$. The angular widths of $A_{m,0}^n$ and $A_{m,1}^n$ are $l_{[n-1]}(\theta_m) - l_{[n]}(\theta_m)$ and $l_{[n]}(\theta_m)$ respectively.

For $m \geq 0$ and $j \in \{0, 1\}$, the first time $k \geq 1$ for which $r_{\theta_m}^{-k}(A_{m,j}^1)$ intersects the interior of S_m^1 is $k = b_m + j\varepsilon_m$. In fact, $r_{\theta_m}^{-b_m - j\varepsilon_m}(A_{m,j}^1)$ is contained in S_m^1 . In general, for $n \geq 1$, the level n triangulation Δ_m^n consists of sectors with angular widths $l_{[n]}(\theta_m)$ and $l_{[n-1]}(\theta_m) - l_{[n]}(\theta_m)$. It is defined inductively in increasing n as follows. Suppose Δ_m^{n-1} has been defined for all $m \geq 0$. Pull back Δ_{m+1}^{n-1} under ψ_m to obtain a triangulation $\check{\Delta}_m^n$ of S_m^1 . For every sector $B \in \check{\Delta}_m^n$, denote $j_B \in \{0, 1\}$ such that B is contained in A_{m,j_B}^1 . Then,

$$\Delta_{m,n} := \{r_{\theta_m}^{-k}B\}_{B \in \check{\Delta}_m^n, k=0,1,\dots,b_m+j_B\varepsilon_m-1}.$$

Proof. Item (1) follows directly from Proposition 2.2 (3_n). Let us prove (2).

Observe that the triangulation Δ_0^n is obtained out of cutting $\overline{\mathbb{D}}$ along the radial arcs $I_0^0, I_0^{-1}, \dots, I_0^{-N+1}$ where $N = N(n)$ is the first positive integer such that I_0^{-N} intersects the interior of the sector S_0^n . It is clear that the intersection of $A_{0,0}^n$ and $A_{0,1}^n$ is I_0^0 . For $j \in \{0, 1\}$, we just need to find the value of $k_j \in \{1, 2, \dots, N-1\}$ such that $A_{0,j}^n$ is bounded by I_0^0 and $I_0^{-k_j}$. Since $A_{0,1}^n$ has angular width equal to $l_{[n]} = l_{[n]}(\theta)$, then by (1), k_1 has to be equal to $q_{[n]}$. By (1) again, $r_\theta^{\varepsilon_n \varepsilon_{n+1} q_{[n-1]}}$ is the rotation by angle $\varepsilon_{n+1}l_{[n-1]}$ which has the same sign as $r_\theta^{[n]}$. As such, $r_\theta^{q_{[n]} - \varepsilon_n \varepsilon_{n+1} q_{[n-1]}}$ rotates by $l_{[n-1]} - l_{[n]}$ in the direction opposite to $r_\theta^{[n]}$. This implies that $k_0 = q_{[n]} - \varepsilon_n \varepsilon_{n+1} q_{[n-1]}$.

The inequality in (3) follows from (1) and the fact that $|\theta_n| < \frac{1}{2}$. We will use (2) and the triangulation Δ_0^n to prove the rest of (3). From the previous paragraph, we have $N \geq q_{[n]}$. Let $A_{0,2}^n \in \Delta_0^n$ be the sector next to $A_{0,1}^n$ that is not equal to

$A_{0,0}^n$, and let $A_{0,3}^n \in \Delta_0^n$ be the sector next to $A_{0,2}^n$ that is not equal to $A_{0,1}^n$. The argument is split into two cases.

- Suppose $\varepsilon_n \varepsilon_{n+1} = +1$. Then, $A_{0,2}^n$ is bounded by $I_0^{-q_{[n]}}$ and $I_0^{-q_{[n-1]}}$ and its angular length is $l_{[n-1]} - l_{[n]}$. For any integer k with $1 \leq k < q_{[n]}$, the radial segment I_0^{-k} is disjoint from the interior of $S_0^n \cup A_{0,2}^n$ which implies that the angular distance between I_0^{-k} and I_0^0 is bounded from below by $l_{[n-1]} - l_{[n]} > \frac{1}{2}l_{[n]}$.
- Suppose $\varepsilon_n \varepsilon_{n+1} = -1$. Then, $A_{0,3}^n$ is bounded by $I_0^{-q_{[n]}}$ and $I_0^{-2q_{[n]}}$ and $A_{0,3}^n$ is bounded by $I_0^{-q_{[n]}+q_{[n-1]}}$. In this case, for any integer k with $1 \leq k < q_{[n]}$, the radial segment I_0^{-k} is disjoint from the interior of $S_0^n \cup A_{0,2}^n \cup A_{0,3}^n$ which implies that the angular distance between I_0^{-k} and I_0^0 is again at least $l_{[n-1]} - l_{[n]} > \frac{1}{2}l_{[n]}$.

In both cases, we have proven the optimality of $q_{[n]}$ described in item (3). $\square$

From the proposition above, for any $n \geq 1$, the first return time of $\mathbf{r}_\theta$ at any point in the interior of S_0^n is either $q_{[n]}$ or $q_{[n]} - \varepsilon_n \varepsilon_{n+1} q_{[n-1]}$. In other words, $\mathbf{r}_\theta^{[n]}|_{S_0^n}$ is the n^{th} pre-renormalization of $\mathbf{r}_\theta$:

$$\mathbf{r}_{\theta_n} = \mathbf{r}_\theta^{[n]}|_{S_0^n} / \mathbf{r}_\theta^{[n-1]}.$$

The gluing map $\psi_{n-1} \circ \dots \circ \psi_0$ projects the map $\mathbf{r}_\theta^{[n]}|_{S_0^n}$, modulo $\mathbf{r}_\theta^{[n-1]}$, to the rotation $\mathbf{r}_{\theta_n}$.

3. DIOPHANTINE APPROXIMATION

In this section, we explore the arithmetic properties of irrationals relative to the modified Gauss map $\mathbf{g} : \Theta \rightarrow \Theta$ defined in the previous section. Most of the results in this section are elementary and similar to those in the study of regular continued fractions. (E.g. compare with [Khi97].) Experts in Diophantine approximation should be familiar with most of the content in this section. We provide details for the sake of completeness.

3.1. The convergents. Given a finite sequence of non-zero integers $x_1, x_2, \dots, x_n$, we will denote

$$[x_1, x_2, \dots, x_n]_- := 1/(x_1 - 1/(x_2 - \dots - 1/(x_{n-1} - 1/x_n) \dots)).$$

Example 3.1. The golden mean irrational

$$\theta_{\text{gm}} = \frac{3 - \sqrt{5}}{2} = 0.381966 \dots$$

is a fixed point of the map $x \mapsto \frac{1}{3-x}$ and so it can be written as

$$\theta_{\text{gm}} = [3, 3, 3, 3, 3, \dots]_-.$$

Lemma 3.2. Consider $\sigma = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$ and the associated sequences of integers $b_n = b_n(\sigma)$, $p_{[n]} = p_{[n]}(\sigma)$ and $q_{[n]} = q_{[n]}(\sigma)$ described in §2.2. For all $n \geq 1$, we have

$$\begin{aligned} (1_n) \quad & p_{[n]} q_{[n-1]} - p_{[n-1]} q_{[n]} = \varepsilon_n, \\ (2_n) \quad & \frac{p_{[n]}}{q_{[n]}} = \sum_{k=1}^n \frac{\varepsilon_k}{q_{[k-1]} q_{[k]}}, \end{aligned}$$

$$(3_n) \quad [\varepsilon_1 b_1, \varepsilon_2 b_2, \dots, \varepsilon_n b_n]_- = \frac{p_{[n]}}{q_{[n]}},$$

$$(4_n) \quad -\frac{1}{2} \leq \frac{p_{[n]}}{q_{[n]}} \leq \frac{1}{2},$$

$$(5_n) \quad q_{[n-1]} < (1 - \theta_{\text{gm}}) q_{[n]}.$$

Proof. For $n = 1$, (1_1) holds since $p_{[0]} = 0$, $q_{[0]} = 1$, and $p_{[1]} = \varepsilon_1$. In general, the recurrence relations (2.1) and (1_{n-1}) imply (1_n) as shown below.

$$\begin{aligned} & p_{[n]} q_{[n-1]} - p_{[n-1]} q_{[n]} \\ &= (b_n p_{[n-1]} - \varepsilon_n \varepsilon_{n-1} p_{[n-2]}) q_{[n-1]} - p_{[n-1]} (b_n q_{[n-1]} - \varepsilon_n \varepsilon_{n-1} q_{[n-2]}) \\ &= \varepsilon_n \varepsilon_{n-1} (p_{[n-1]} q_{[n-2]} - p_{[n-2]} q_{[n-1]}) \\ &= \varepsilon_n \varepsilon_{n-1} \varepsilon_{n-1} = \varepsilon_n. \end{aligned}$$

Let us prove (2_n) . The base case (2_n) is clear. For $n \geq 2$, (2_n) follows from (4_{n-1}) and (1_n) since

$$\frac{p_{[n]}}{q_{[n]}} - \frac{p_{[n-1]}}{q_{[n-1]}} = \frac{p_{[n]} q_{[n-1]} - p_{[n-1]} q_{[n]}}{q_{[n-1]} q_{[n]}} = \frac{\varepsilon_n}{q_{[n-1]} q_{[n]}}.$$

By elementary calculation, we have (3_1) and (3_2) :

$$[\varepsilon_1 b_1]_- = \frac{\varepsilon_1}{b_1} = \frac{p_{[1]}}{q_{[1]}}, \quad [\varepsilon_1 b_1, \varepsilon_2 b_2]_- = \frac{\varepsilon_1 b_2}{b_2 b_1 - \varepsilon_2 \varepsilon_1} = \frac{p_{[2]}}{q_{[2]}}.$$

Suppose we have (3_n) for all $n \in \{1, \dots, k\}$ for some integer $k \geq 2$. Let $x = b_k - \frac{\varepsilon_k \varepsilon_{k+1}}{b_{k+1}}$. Then, by the recurrence relations,

$$\begin{aligned} [\varepsilon_1 b_1, \dots, \varepsilon_{k-1} b_{k-1}, \varepsilon_k b_k, \varepsilon_{k+1} b_{k+1}]_- &= [\varepsilon_1 b_1, \dots, \varepsilon_{k-1} b_{k-1}, \varepsilon_k x]_- \\ &= \frac{x p_{[k-1]} - \varepsilon_{k-1} \varepsilon_k p_{[k-2]}}{x q_{[k-1]} - \varepsilon_{k-1} \varepsilon_k q_{[k-2]}} \\ &= \frac{p_{[k]} - \varepsilon_k \varepsilon_{k+1} p_{[k-1]} / b_{k+1}}{q_{[k]} - \varepsilon_k \varepsilon_{k+1} q_{[k-1]} / b_{k+1}} = \frac{p_{[k+1]}}{q_{[k+1]}}. \end{aligned}$$

Hence, (3_n) holds for all n .

Next, we will use (3_n) to prove (4_n) . The base case (4_1) is clear. Suppose (4_{n-1}) holds for some $n \geq 2$. In particular, $y := \varepsilon_2 [\varepsilon_2 b_2, \dots, \varepsilon_n b_n]_-$ is contained in the interval $[0, \frac{1}{2}]$. Then, we can write

$$\left| \frac{p_{[n]}}{q_{[n]}} \right| = \left| \frac{1}{\varepsilon_1 b_1 - \varepsilon_2 y} \right| = \frac{1}{b_1 - \varepsilon_1 \varepsilon_2 y} = \begin{cases} \frac{1}{\bar{a}_1 + 1 - y} & \text{if } \varepsilon_1 \varepsilon_2 = +1, \\ \frac{1}{\bar{a}_1 + y} & \text{if } \varepsilon_1 \varepsilon_2 = -1. \end{cases}$$

Since $\bar{a}_1 \geq 2$ and $0 \leq y \leq 2$, the equation above gives us (4_n) .

To prove (5_n) , we will show that for all $n \geq 1$, we have

$$(\spadesuit_n) \quad \theta_{\text{gm}} - 1 < \varepsilon_n \varepsilon_{n+1} \frac{q_{[n-1]}}{q_{[n]}} < \theta_{\text{gm}}.$$

The base case $(\spadesuit_1)$ follows from $\frac{1}{3} < \theta_{\text{gm}} < \frac{1}{2} < 1 - \theta_{\text{gm}}$ and the following observation:

$$\varepsilon_1 \varepsilon_2 \frac{q_{[0]}}{q_{[1]}} = \frac{\varepsilon_1 \varepsilon_2}{b_1} = \begin{cases} \frac{1}{\bar{a}_1 + 1} \in (0, \frac{1}{3}] & \text{if } \varepsilon_1 \varepsilon_2 = +1, \\ -\frac{1}{\bar{a}_1} \in [-\frac{1}{2}, 0) & \text{if } \varepsilon_1 \varepsilon_2 = -1. \end{cases}$$

In general, for $n \geq 2$, the recurrence relation 2.1 gives us:

$$\varepsilon_n \varepsilon_{n+1} \frac{q_{[n-1]}}{q_{[n]}} = \frac{\varepsilon_n \varepsilon_{n+1}}{b_n - \varepsilon_n \varepsilon_{n-1} \frac{q_{[n-2]}}{q_{[n-1]}}} = \begin{cases} \frac{1}{\bar{a}_n + 1 - \varepsilon_n \varepsilon_{n-1} \frac{q_{[n-2]}}{q_{[n-1]}}} & \text{if } \varepsilon_n \varepsilon_{n+1} = +1, \\ -\frac{1}{\bar{a}_n - \varepsilon_n \varepsilon_{n-1} \frac{q_{[n-2]}}{q_{[n-1]}}} & \text{if } \varepsilon_n \varepsilon_{n+1} = -1. \end{cases}$$

Then, $(\spadesuit_n)$ follows from $(\spadesuit_{n-1})$:

$$\varepsilon_n \varepsilon_{n+1} \frac{q_{[n-1]}}{q_{[n]}} \in \begin{cases} \left(0, \frac{1}{3-\theta_{\text{gm}}}\right) = (0, \theta_{\text{gm}}) & \text{if } \varepsilon_n \varepsilon_{n+1} = +1, \\ \left(-\frac{1}{2-\theta_{\text{gm}}}, 0\right) = (\theta_{\text{gm}} - 1, 0) & \text{if } \varepsilon_n \varepsilon_{n+1} = -1. \end{cases} \quad \square$$

Proposition 3.3. Consider $\sigma = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$ and the associated sequences $b_n = b_n(\sigma)$, $p_{[n]} = p_{[n]}(\sigma)$, and $q_{[n]} = q_{[n]}(\sigma)$. The limit $\mathfrak{X}(\sigma) = \lim_{n \rightarrow \infty} \frac{p_{[n]}}{q_{[n]}}$ converges to an irrational in Θ equal to

$$\mathfrak{X}(\sigma) = \sum_{k=1}^{\infty} \frac{\varepsilon_k}{q_{[k-1]} q_{[k]}}$$

Moreover, for every $n \geq 1$,

$$\frac{\theta_{\text{gm}}}{q_{[n]} q_{[n+1]}} \leq \left| \mathfrak{X}(\sigma) - \frac{p_{[n]}}{q_{[n]}} \right| \leq \frac{2 - \theta_{\text{gm}}}{q_{[n]} q_{[n+1]}}.$$

Proof. For $k \geq 1$, denote $x_k = \frac{\varepsilon_k}{q_{[k-1]} q_{[k]}}$. By Lemma 3.2 (5), we have

$$(3.1) \quad \left| \frac{x_k}{x_{k+1}} \right| = \frac{q_{[k-1]}}{q_{[k+1]}} < (1 - \theta_{\text{gm}})^2 = \theta_{\text{gm}} < 1.$$

Therefore, the series $\sum_{k \geq 1} x_k$ converges to an irrational number θ . By items (2), (3), and (4) of Lemma 3.2, θ is in Θ , is equal to $\mathfrak{X}(\sigma)$, and has an infinite modified continued fraction expansion $\theta = [\varepsilon_1 b_1, \varepsilon_2 b_2, \varepsilon_3 b_3, \dots]_-$. Moreover, for $n \geq 1$, the inequality (3.1) gives us

$$\left| \theta - \frac{p_{[n]}}{q_{[n]}} \right| = \left| \sum_{k=n+1}^{\infty} x_k \right| \leq \frac{|x_{n+1}|}{1 - \theta_{\text{gm}}} = \frac{1}{(1 - \theta_{\text{gm}}) q_{[n]} q_{[n+1]}} = \frac{2 - \theta_{\text{gm}}}{q_{[n]} q_{[n+1]}}$$

and

$$\begin{aligned} \left| \theta - \frac{p_{[n]}}{q_{[n]}} \right| &\geq |x_{n+1}| - \left| \sum_{k=n+2}^{\infty} x_k \right| \geq \frac{1}{q_{[n]} q_{[n+1]}} - \frac{2 - \theta_{\text{gm}}}{q_{[n+1]} q_{[n+2]}} \\ &\geq \frac{1 - (2 - \theta_{\text{gm}})(1 - \theta_{\text{gm}})^2}{q_{[n]} q_{[n+1]}} = \frac{\theta_{\text{gm}}}{q_{[n]} q_{[n+1]}}. \end{aligned}$$

Note that we have used Lemma 3.2 (5) on the last line above. $\square$

The proposition above gives us a well-defined function

$$\mathfrak{X} : \Sigma^{\mathbb{N}} \rightarrow \Theta.$$

Recall the functions $\mathfrak{g} : \Theta \rightarrow \Theta$ and $\mathfrak{y} : \Theta \rightarrow \Sigma^{\mathbb{N}}$ defined in the previous section.

Theorem 3.4. *The function $\mathfrak{X} : \Sigma^{\mathbb{N}} \rightarrow \Theta$ is a homeomorphism with inverse $\mathfrak{Y} : \Theta \rightarrow \Sigma^{\mathbb{N}}$. Moreover, $\mathfrak{X}$ is a conjugacy between the standard shift map $\mathfrak{s} : \Sigma^{\mathbb{N}} \rightarrow \Sigma^{\mathbb{N}}$ and the map $\mathfrak{g} : \Theta \rightarrow \Theta$.*

This theorem settles Theorem A.

Proof. Let $\sigma = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$ and let $b_n = b_n(\sigma)$ for $n \geq 1$. We have

$$\mathfrak{X}(\mathfrak{s}(\sigma)) = [\varepsilon_2 b_2, \varepsilon_3 b_3, \dots]_- = -\frac{1}{\mathfrak{X}(\sigma)} + \varepsilon_1 b_1 = \mathfrak{g}(\mathfrak{X}(\sigma)).$$

Let $\theta = \mathfrak{X}(\sigma)$. Then, $\varepsilon(\theta) = \vartheta_1$ and $\bar{a}(\theta) = \bar{a}_1$. Since $\mathfrak{g}^{n-1}(\theta) = \mathfrak{X}(\mathfrak{s}^{n-1}(\sigma))$, we also have that $\varepsilon(\mathfrak{g}^{n-1}(\theta)) = \vartheta_n$ and $\bar{a}(\mathfrak{g}^{n-1}(\theta)) = \bar{a}_n$. Therefore, $\mathfrak{Y} \circ \mathfrak{X}$ is equal to the identity map on $\Sigma^{\mathbb{N}}$.

We have established that $\mathfrak{Y}$ is surjective. Now, we will show the injectivity of $\mathfrak{Y}$. Fix $\sigma = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}$. For $k \geq 1$, denote

$$E_k(\sigma) = \{ \theta \in \Theta : \varepsilon(\mathfrak{g}^{n-1}(\theta)) = \varepsilon_n \text{ and } \bar{a}(\mathfrak{g}^{n-1}(\theta)) = \bar{a}_n \text{ for } 1 \leq n \leq k \}.$$

The preimage $\mathfrak{Y}^{-1}(\sigma)$ is non-empty and it is contained in $\cap_{n \geq 1} E_k(\sigma)$. The set $E_1(\sigma)$ is the open subset of Θ consisting of irrationals between $\frac{1}{\varepsilon_1 \bar{a}_1}$ and $\frac{1}{\varepsilon_1(\bar{a}_1+1)}$; since $\bar{a} \geq 2$, this open set has Euclidean length bounded above by 6^{-1} . In general, $\mathfrak{g}^{k-1}$ sends $E_k(\sigma)$ homeomorphically onto $E_1(\mathfrak{s}^{k-1}(\sigma))$. Observe that $\mathfrak{g}$ is uniformly expanding: $\mathfrak{g}' \geq 4$. This implies that $E_k(\sigma)$ is equal to $\Theta \cap J$ for some open interval J of Euclidean length at most $6^{-1} \cdot 4^{1-k}$. In particular, $\mathfrak{Y}^{-1}(\sigma)$ is a singleton and thus $\mathfrak{Y}$ is injective.

Pick $\sigma \in \Sigma^{\mathbb{N}}$ and let $\theta = \mathfrak{X}(\sigma)$. The sets $E_k(\sigma)$, $k \geq 1$ described above form a basis of open neighborhoods of θ . The images $\mathfrak{Y}(E_k(\sigma))$, $k \geq 1$ are precisely the cylinder sets about σ which form a basis of open neighborhoods of σ with respect to the topology of $\Sigma^{\mathbb{N}}$. Therefore, both $\mathfrak{X}$ and $\mathfrak{Y}$ are continuous. $\square$

Example 3.5. For the golden mean irrationals in Θ are $\theta_{\text{gm}} = \frac{3-\sqrt{5}}{2}$ and $-\theta_{\text{gm}} = \frac{\sqrt{5}-3}{2}$, we have

$$\begin{aligned} \mathfrak{Y}(\theta_{\text{gm}}) &= \langle (+, 2), (+, 2), (+, 2), (+, 2), \dots \rangle, \\ \mathfrak{Y}(-\theta_{\text{gm}}) &= \langle (-, 2), (-, 2), (-, 2), (-, 2), \dots \rangle. \end{aligned}$$

The corresponding modified continued fraction expansions are

$$\theta_{\text{gm}} = [3, 3, 3, \dots]_- = \frac{1}{3 - \frac{1}{3 - \frac{1}{3 - \dots}}}, \quad -\theta_{\text{gm}} = [-3, -3, -3, \dots]_- = \frac{1}{-3 - \frac{1}{-3 - \frac{1}{-3 - \dots}}}.$$

Both θ_{gm} and $-\theta_{\text{gm}}$ have the same first return times, namely

$$q_{[0]} = 1, \quad q_{[1]} = 3, \quad q_{[2]} = 8, \quad q_{[3]} = 21, \quad q_{[4]} = 55, \quad q_{[5]} = 144, \quad \dots$$

Each $q_{[n]}$ is the $(2n+2)^{\text{th}}$ Fibonacci number.

3.2. The nearest integer continued fraction expansion. Consider the nearest integer continued fraction map

$$f_{1/2} : \Theta \rightarrow \Theta, \quad f_{1/2}(\theta) = \frac{1}{|\theta|} - \left\lfloor \frac{1}{|\theta|} + \frac{1}{2} \right\rfloor.$$

For $n \geq 0$, we define the sequences $\{\tilde{a}_n\}_{n \geq 1}$ in $\mathbb{N}_{\geq 2}$, $\{\tilde{\theta}_n\}_{n \geq 0}$ in $(0, \frac{1}{2}) \setminus \mathbb{Q}$, and $\{\tilde{\varepsilon}_n\}_{n \geq 1}$ in $\{-1, +1\}$ inductively as follows:

- $\tilde{\varepsilon}_{n+1}$ is the sign of $\tilde{\theta}_n$;
- $\tilde{a}_{n+1} = \left\lfloor |\tilde{\theta}_n|^{-1} + \frac{1}{2} \right\rfloor$ is the integer closest to $|\tilde{\theta}_n|^{-1}$;
- $\tilde{\theta}_{n+1} = f_{1/2}(\tilde{\theta}_n) = |\tilde{\theta}_n|^{-1} - \tilde{a}_{n+1}$.

Then, we can write

$$(\ddagger) \quad \theta = \frac{\tilde{\varepsilon}_1}{\tilde{a}_1 + \frac{\tilde{\varepsilon}_2}{\tilde{a}_2 + \frac{\tilde{\varepsilon}_3}{\tilde{a}_3 + \dots}}}.$$

The map $f_{1/2}$ is an even function and it is related to $\mathbf{g}$ by

$$f_{1/2}(\theta) = \begin{cases} \mathbf{g}(\theta) & \text{if } \theta < 0, \\ -\mathbf{g}(\theta) & \text{if } \theta > 0. \end{cases}$$

It produces the same convergents $\tilde{p}_{[n]}/\tilde{q}_{[n]}$ as those $p_{[n]}/q_{[n]}$ of $\mathbf{g}$ up to a sign change. The terms $\tilde{a}_n$'s coincide with the b_n 's described in Section 2.2. Our modified Gauss map $\mathbf{g}$ has two primary advantages over $f_{1/2}$.

- The map $\mathbf{g}$ locally preserves orientation everywhere, whereas $f_{1/2}$ reverses orientation the positive irrationals.
- While $f_{1/2}$ still acts as a shift on $(\tilde{\varepsilon}_n, \tilde{a}_n)$'s, these $\tilde{a}_n$'s satisfy the awkward admissibility rule that $\tilde{a}_n \geq 3$ whenever $\tilde{\varepsilon}_n \tilde{\varepsilon}_{n+1} = -1$.

Nearest integer continued fraction was initially introduced by Bernhard Minnigerode [Min73] in 1873. The notation $f_{1/2}$ was adapted from Nakada [Nak81] who introduced the one-parameter family

$$f_\alpha : [\alpha - 1, \alpha) \rightarrow [\alpha - 1, \alpha), \quad \alpha \in [\tfrac{1}{2}, 1]$$

of continued fractions map, where f_1 is the classical Gauss map.

In the field of complex dynamics, the expansion $(\ddagger)$ first appeared in the seminal work of Yoccoz [Yoc95] in which he introduced sector renormalization. The adaptation of $(\ddagger)$ was also evident in the work of Inou-Shishikura [IS06] on near-parabolic renormalization as well as its numerous applications, for example [CC15, AC17, Che25]. In those works, the authors are primarily working with the sequence of positive irrationals

$$\alpha_n = |\tilde{\theta}_n|, \quad n \geq 0.$$

3.3. The regular continued fraction expansion. Every irrational $\theta \in \Theta$ admits a unique continued fraction expansion

$$\theta = \varepsilon [a_1, a_2, a_3, \dots]_+ := \frac{\varepsilon}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$$

where $\varepsilon \in \{-1, +1\}$, $a_1 \in \mathbb{N}_{\geq 2}$, and $a_n \in \mathbb{N}$ for all $n \geq 2$. For $n \geq 1$, the n^{th} rational approximation of θ is

$$\frac{p_n}{q_n} = \varepsilon [a_1, a_2, \dots, a_n]_+$$

where p_n and q_n are co-prime integers and $q_n > 0$. These p_n 's and q_n 's start with

$$p_n = 0, q_0 = 1, \quad p_1 = \varepsilon, q_1 = a_1,$$

and the rest follow the recurrence relation

$$p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2} \quad \text{for } n \geq 2.$$

Note that the q_n 's are independent of the sign ε . The numbers $q_n \theta - p_n$ alternate in sign, that is

$$(q_n \theta - p_n)(q_{n+1} \theta - p_{n+1}) < 0 \quad \text{for all } n \geq 0.$$

There are two advantages of the regular continued fraction expansion, namely the simpler recursive formula for the rational approximations and the alternating nature of points of closest return. The main disadvantage is that the regular expansion does not get transformed as nicely under sector renormalization:

$$\mathfrak{g}(\varepsilon[a_1, a_2, \dots]_+) = \begin{cases} -\varepsilon[a_2, a_3, \dots]_+ & \text{if } a_2 \geq 2, \\ \varepsilon[a_3 + 1, a_4, \dots]_+ & \text{if } a_2 = 1. \end{cases}$$

Arithmetically, the regular expansion is more suited to studying the action of the traditional Gauss map $G(\theta) = \{\frac{1}{\theta}\}$ acting on irrationals on the unit interval $(0, 1)$.

There is a singularization procedure to convert the regular continued fraction expansion $\theta = \varepsilon[a_1, a_2, \dots]_+$ to the modified expansion with terms governed by $\langle(\varepsilon_n, \bar{a}_n)\rangle_{n \geq 1} \in \Sigma^{\mathbb{N}}$. First, set $\varepsilon_1 = \varepsilon$ and $\bar{a}_1 = a_1$. To get $(\varepsilon_2, \bar{a}_2)$, there are two cases.

- If $a_2 \geq 2$, then $\varepsilon_2 = -\varepsilon_1$, and $\bar{a}_2 = a_2$.
- Else if $a_2 = 1$, then $\varepsilon_2 = \varepsilon_1$, and $\bar{a}_2 = a_3 + 1$.

In general, it is convenient to use the increasing sequence of integers

$$0 = t_0 < t_1 < t_2 < t_3 < \dots$$

defined as follows. For $n \geq 0$,

- if $a_{t_n+2} \geq 2$, then $t_{n+1} = t_n + 1$, $\varepsilon_{n+2} = -\varepsilon_{n+1}$ and $\bar{a}_{n+2} = a_{t_n+2}$;
- if $a_{t_n+2} = 1$, then $t_{n+1} = t_n + 2$, $\varepsilon_{n+2} = \varepsilon_{n+1}$ and $\bar{a}_{n+2} = a_{t_n+3} + 1$.

With this convention, we have that

$$q_{[n]} = q_{t_n} \quad \text{for all } n \geq 0.$$

Geometrically, here is what happens. Consider the lengths

$$l_n := |q_n \theta - p_n|, \quad n \geq 1$$

which form strictly decreasing sequence. The rate of decrease is generally non-uniform and the ratio l_{n+1}/l_n can be arbitrarily close to 1. According to Proposition 2.3, the subsequence $\{t_n\}_{n \geq 0}$ records the moments in which we have definite contraction by one half:

$$l_{t_{n+1}} < \frac{l_{t_n}}{2}.$$

The moments that are not of the form t_n are deemed as “negligible” levels and can be thrown out.

4. THE COMPACTIFICATION

In this section, we will discuss a natural compactification of Θ that respects the dynamical behavior of $\mathfrak{g}$.

4.1. The compactification $\bar{\Theta}$.

Definition 4.1. Consider a continuous dynamical system $f : X \rightarrow X$ on a topological space X . A *compactification* of the dynamical system (X, f) is a triple $(j, \bar{X}, \bar{f})$ where

- (a) $j : X \rightarrow \bar{X}$ is an embedding with dense image,
- (b) $\bar{X}$ is a compact space, and
- (c) $\bar{f} : \bar{X} \rightarrow \bar{X}$ is a continuous map such that $\bar{f} \circ j = j \circ f$.

We say that a compactification $(j, \bar{X}, \bar{f})$ of (X, f) *respects* an embedding $h : X \rightarrow Y$ onto a compact space Y if there exists a continuous map $\bar{h} : \bar{X} \rightarrow Y$, called a *j-extension* of h , such that $\bar{h} \circ j = h$.

In the definition above, both $\bar{f}$ and $\bar{h}$ are unique because of the density of $j(X)$ in $\bar{X}$. Theorem A gives us a natural compactification of $(\Theta, \mathfrak{g})$.

Definition 4.2. Let $\bar{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, where the addition of the point $\infty = \lim_{n \rightarrow \infty} n$ makes it a compactification of the discrete space $\mathbb{N}$. Let

$$\bar{\Sigma} := \{-1, +1\} \times \bar{\mathbb{N}}_{\geq 2} \quad \text{and} \quad \bar{\Theta} := \bar{\Sigma}^{\mathbb{N}}.$$

Given an element $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}$ of $\bar{\Theta}$, any integer $n \geq 0$ such that $\bar{a}_{n+1} = \infty$ is called a *parabolic level* for θ . We say that θ is called an *enriched rational* if parabolic levels exist, an *irrational* if otherwise. For $n \geq 1$, we will denote

$$b_n(\theta) := \begin{cases} \bar{a}_n + \frac{1 + \varepsilon_n \varepsilon_{n+1}}{2} & \text{if } \bar{a}_n < \infty, \\ \infty & \text{if } \bar{a}_n = \infty. \end{cases}$$

The sequence space $\bar{\Theta}$ is topologically a Cantor set. The inclusion map $\iota : \Sigma^{\mathbb{N}} \rightarrow \bar{\Theta}$ induces a dense embedding

$$\bar{\iota} := \iota \circ \mathfrak{J} : \Theta \rightarrow \bar{\Theta}$$

and the modified Gauss map $\mathfrak{g}$ naturally admits an $\bar{\iota}$ -extension

$$\bar{\mathfrak{g}} : \bar{\Theta} \rightarrow \bar{\Theta}$$

given by the standard shift map $\bar{\mathfrak{g}} \equiv \mathfrak{s}$. Hence, $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$ is a compactification of $(\Theta, \mathfrak{g})$.

There are three compact embeddings of Θ of high interest, each of which admits an $\bar{\iota}$ -extension.

4.1.1. *The modular embedding.* The first one is the naive embedding

$$\mu : \Theta \rightarrow \mathbb{T}, \quad \theta \mapsto \theta \pmod{1}$$

into the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, called the *modular embedding*.

Proposition 4.3. *The modular embedding μ admits an $\bar{\iota}$ -extension $\bar{\mu} : \bar{\Theta} \rightarrow \mathbb{T}$. For every rational number θ in $\mathbb{T}$, $\bar{\mu}^{-1}(\theta)$ is homeomorphic to $\bar{\Theta}$ itself.*

Proof. For an enriched rational $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \bar{\Theta}$ with smallest parabolic level $N \geq 0$, we set

$$\bar{\mu}(\theta) = \begin{cases} 0 & \text{if } N = 0, \\ [\varepsilon_1 b_1, \dots, \varepsilon_N b_N]_- & \text{if } N \geq 1. \end{cases}$$

The continuity of $\bar{\mu} : \bar{\Theta} \rightarrow \mathbb{T}$ follows from the proof of Proposition 3.4.

For the second claim, observe that

$$\bar{\mu}^{-1}(0) = \{\bar{a}_1 = \infty\}, \quad \bar{\mu}^{-1}\left(\pm \frac{1}{2}\right) = \{\varepsilon_1 \varepsilon_2 = -, \bar{a}_1 = 2, \bar{a}_2 = \infty\},$$

and for $m \geq 3$ and $\varepsilon \in \{-1, +1\}$, we have

$$\bar{\mu}^{-1}\left(\frac{\varepsilon}{m}\right) = \left\{ \varepsilon_1 = \varepsilon, \bar{a}_1 + \frac{1 + \varepsilon \varepsilon_2}{2} = m, \bar{a}_2 = \infty \right\}.$$

All of the above are homeomorphic to $\bar{\Theta}$. For other rational numbers θ , let $k \geq 2$ be the first moment such that $\mathbf{g}(\theta) = 0$, then $\varepsilon_j(\theta)$ and $\bar{a}_j(\theta)$ are well-defined and finite for $j \leq k-1$, $\bar{a}_{k+1} = \infty$, the sequences $\{\varepsilon_j\}_{j \geq k+1}$ and $\{\bar{a}_j\}_{j \geq k+2}$ are completely arbitrary, and the pair $(\varepsilon_k, \bar{a}_k)$ is completely determined by ε_{k+1} in the way that is similar to the $\pm \frac{1}{m}$ case. $\square$

This proposition tells us that, with respect to $\bar{\mu}$, the space $\bar{\Theta}$ can be viewed as a way of blowing up the circle $\mathbb{T}$ at its rational points. In light of applications in holomorphic dynamics, the structure of the fibers of a rational point on the circle sheds light on the necessity to blow up the Mandelbrot set at parabolic parameters in order to claim continuity of dynamical sets (e.g. the Mother Hedgehog, the (filled) Julia set, etc).

4.1.2. *The segment embedding.* The second embedding to consider is

$$\eta : \Theta \rightarrow [0, 1], \quad \eta(\theta) = \begin{cases} \theta & \text{if } \theta > 0, \\ \theta + 1 & \text{if } \theta < 0, \end{cases}$$

which we call the *segment embedding*. This map is more natural to consider rather than μ for the following reason. The modified Gauss map $\mathbf{g} : \Theta \rightarrow \Theta$ extends to a self-map of $\mathbb{T}$ that is continuous (in fact, C^1 smooth) everywhere except at the origin. See Figure 3 for reference. Since the origin is singular from both the left and right, it is natural to distinguish between the left side of 0 and the right side of 0. The segment embedding precisely does this as it identifies the ends $-\frac{1}{2}$ and $\frac{1}{2}$ and splits the left and right sides of the origin to two separate ends 0 and 1. The diagram below is commutative.

$$\begin{array}{ccc} \Theta & \xrightarrow{\eta} & [0, 1] \\ & \searrow \mu & \downarrow 0 \sim 1 \\ & & \mathbb{T} \end{array}$$

Proposition 4.4. *The segment embedding η admits an $\bar{\iota}$ -extension $\bar{\eta} : \bar{\Theta} \rightarrow [0, 1]$, which is surjective.*

Proof. On the irrationals, η can be symbolically written as

$$\eta \circ \mathfrak{X}(\langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}) = \frac{1 - \varepsilon_1}{2} + [\varepsilon_1 b_1, \varepsilon_2 b_2, \dots]_-.$$

Define the extension $\bar{\eta}$ symbolically in the same way as $\eta \circ \mathfrak{X}$. For the enriched rationals, we make the convention that if $N \geq 0$ is the smallest parabolic level, then

$$[\varepsilon_1 b_1, \varepsilon_2 b_2, \dots]_- = \begin{cases} 0 & \text{if } N = 0, \\ [\varepsilon_1 b_1, \varepsilon_2 b_2, \dots, \varepsilon_N b_N]_- & \text{if } N \geq 1. \end{cases}$$

The continuity and surjectivity of $\bar{\eta}$ are straightforward. $\square$

4.1.3. *The dynamical embedding.* The third embedding to consider is the *dynamical embedding* $\text{ord} : \Theta \rightarrow \Omega$ mentioned in the introduction. It has a strong motivation arising from the renormalization theory of critical circle maps and neutral quadratic polynomials.

Given integers $k, l \geq 1$, let $\Omega_{k,l}$ denote the set of ambient isotopy classes of maps $f : \{0, 1, \dots, k\}^{l+1} \rightarrow \mathbb{T}$, i.e. the set of maps f modulo isotopy of the image of f in $\mathbb{T}$. It corresponds to the set of all circular permutations of $s \in \mathbb{N}$ elements where $1 \leq s \leq ((k+1)^{l+1} - 1)!$. The set $\Omega_{k,l}$ is finite and we will equip it with the discrete topology. We will consider the product space

$$\Omega = \prod_{k,l \geq 1} \Omega_{k,l}$$

which is a Cantor set. Given an irrational $\theta \in \Theta$, we denote by $\text{ord}_{k,l}(\theta) \in \Omega_{k,l}$ the ambient isotopy class of the embedding

$$(j_0, \dots, j_l) \mapsto \left(\mathbb{r}_\theta^{[0]} \right)^{j_0} \circ \left(\mathbb{r}_\theta^{[1]} \right)^{j_1} \circ \dots \circ \left(\mathbb{r}_\theta^{[l]} \right)^{j_l} (1).$$

This gives us a locally constant map

$$\text{ord}_{k,l} : \Theta \rightarrow \Omega_{k,l} \quad \text{for all } k, l \geq 1,$$

which altogether induces the embedding

$$\text{ord} : \Theta \rightarrow \Omega, \quad \text{ord}(\theta) = \{\text{ord}_{k,l}(\theta)\}_{k,l}.$$

Proposition 4.5. *The embedding ord admits an $\bar{\iota}$ -extension $\overline{\text{ord}} : \bar{\Theta} \rightarrow \Omega$, which is again an embedding.*

Proof. The proof is split into two steps, namely the continuity and well-definedness of $\overline{\text{ord}}$, and the injectivity of $\overline{\text{ord}}$. Since $\bar{\Theta}$ is compact and Ω is Hausdorff, these two steps are enough to prove that $\overline{\text{ord}}$ is an embedding.

For $\theta \in \bar{\Theta}$ and $M, l \in \mathbb{N}$, define

$$\bar{U}_{M,l}(\theta) := \left\{ \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \bar{\Theta} : \begin{array}{l} \text{for } n \in \{1, \dots, l\}, \varepsilon_n = \varepsilon_n(\theta), \\ \bar{a}_n \geq M \text{ if } \bar{a}_n = \infty, \text{ and} \\ \bar{a}_n = \bar{a}_n(\theta) \text{ if } \bar{a}_n < \infty \end{array} \right\}.$$

The set $\bar{U}_{M,l}(\theta)$ is a clopen neighborhood of θ in $\bar{\Theta}$. Let $U_{M,l}(\theta)$ be the open subset of Θ given by

$$U_{M,l}(\theta) = \bar{\iota}^{-1}(\bar{U}_{M,l}(\theta)).$$

Denote

$$m_l(\theta) := \max\{\bar{a}_j : \bar{a}_j < \infty, 1 \leq j \leq l\}.$$

Claim: For any k and $l \geq 1$, there exists a sufficiently high positive integer M depending only on k, l , and $m_l(\theta)$ such that $\text{ord}_{k,l}$ is constant in $U_{M,l}(\theta)$. This constant will be denoted by $\overline{\text{ord}}_{k,l}(\theta)$.

Proof. Consider two distinct tuples $(i_s)_{0 \leq s \leq l}$ and $(j_s)_{0 \leq s \leq l}$ in $\{0, \dots, k\}^{l+1}$, and write $w_s = j_s - i_s$. Since we are dealing with irrational numbers $\theta \in U_{M,l}(\theta)$, we conclude that the collision of the $(i_s)_s$ -iterate and the $(j_s)_s$ -iterate can be written arithmetically as

$$\mathbb{r}_\theta^{i_0 q_{[0]}(\theta) + \dots + i_l q_{[l]}(\theta)}(1) = \mathbb{r}_\theta^{j_0 q_{[0]}(\theta) + \dots + j_l q_{[l]}(\theta)}(1) \quad \text{iff} \quad P(\theta) := \sum_{s=0}^m w_s q_{[s]}(\theta) = 0.$$

Our goal is reduced to showing that in $U_{M,l}(\boldsymbol{\theta})$, the function $P(\theta)$ is either non-vanishing or constantly zero. If $\bar{a}_j < \infty$ for all $j \in \{1, \dots, l\}$, then $P(\theta)$ is constant in $U_{M,l}(\boldsymbol{\theta})$ and there is nothing further to be done. So let us suppose otherwise.

Observe that for all $t \in \{1, \dots, l\}$,

$$\left| \sum_{s=0}^{t-1} w_s q_{[s]}(\theta) \right| \leq k \left| \sum_{s=0}^{t-1} q_{[s]}(\theta) \right| \leq k q_{[t]}.$$

To keep track of the indices that give us infinity, we write

$$\{s \in \{1, \dots, l\} : \bar{a}_s = \infty\} = \{t_1, t_2, \dots, t_\nu\}$$

where $t_j < t_{j+1}$. The recurrence relation allows us to write

$$\sum_{s=t_\nu-1}^m w_s q_{[s]}(\theta) = X_\nu q_{[t_\nu]}(\theta) + Y_\nu q_{[t_\nu-1]}(\theta).$$

for some integers X_ν and Y_ν independent of $\theta \in U_{M,l}(\boldsymbol{\theta})$ whose absolute values are bounded above by some constant $C_\nu > 0$ depending only on k , l , and $m_l(\boldsymbol{\theta})$. Observe that

$$\begin{aligned} |P(\theta)| &\geq \left| \sum_{s=t_\nu-1}^m w_s q_{[s]}(\theta) \right| - \left| \sum_{s=0}^{t_\nu-2} w_s q_{[s]}(\theta) \right| \\ &\geq |X_\nu| q_{[t_\nu]}(\theta) - |Y_\nu| q_{[t_\nu-1]}(\theta) - k q_{[t_\nu-1]}(\theta) \\ &\geq |X_\nu| (\bar{a}_{t_\nu}(\theta) - 1) - |Y_\nu| - k q_{[t_\nu-1]}(\theta). \end{aligned}$$

We will demand that $M \geq C_\nu + k + 2$. This would mean that if $U_\nu \neq 0$, then

$$\left| |X_\nu| (\bar{a}_{t_\nu}(\theta) - 1) - |Y_\nu| - k \right| \geq \bar{a}_{t_\nu}(\theta) - 1 - C - k \geq 1,$$

which means that $P(\theta) \neq 0$ for all $\theta \in U_{M,l}(\boldsymbol{\theta})$. Let us suppose otherwise that $X_\nu = 0$. In this case, we can go further down to index $t_{\nu-1}$ and write

$$\sum_{s=t_{\nu-1}-1}^m w_s q_{[s]}(\theta) = X_{\nu-1} q_{[t_{\nu-1}]}(\theta) + Y_{\nu-1} q_{[t_{\nu-1}-1]}(\theta),$$

where again, $X_{\nu-1}$ and $Y_{\nu-1}$ are integers that are independent of $\theta \in U_{M,l}(\boldsymbol{\theta})$ whose absolute values are bounded above by some constant $C_{\nu-1} > 0$ depending only on k , l , and $m_l(\boldsymbol{\theta})$. By demanding again that $M \geq C_{\nu-1} + k + 2$, if $X_{\nu-1} \neq 0$, we have that $P(\theta)$ is non-vanishing, and if otherwise, we can go down to the next index and repeat the argument.

If the inductive argument above survives, we arrive at the situation where

$$\sum_{s=t_1-1}^m w_s q_{[s]}(\theta) = Y_1 q_{[t_1-1]}(\theta)$$

where Y_1 is an integer independent of $\theta \in U_{M,l}(\boldsymbol{\theta})$. In this final case, the whole sum $P(\theta)$ would have to be a constant independent of $\theta \in U_{M,l}(\boldsymbol{\theta})$, and we are done. $\square$

The claim above implies that $\overline{\text{ord}}(\boldsymbol{\theta}) = \{\overline{\text{ord}}_{k,l}(\boldsymbol{\theta})\}_{k,l \geq 1}$ is a well-defined continuous extension of ord . Next, we need to show that $\overline{\text{ord}}$ is injective. Pick two distinct elements $\boldsymbol{\theta} = \langle (\varepsilon_n, \bar{a}_n)_{n \geq 1} \rangle$ and $\boldsymbol{\theta}' = \langle (\varepsilon'_n, \bar{a}'_n)_{n \geq 1} \rangle$ of $\bar{\Theta}$. There are a few cases.

Case 1: $\varepsilon_l \neq \varepsilon'_l$ for some minimal $l \geq 1$.

Let $\theta, \theta' \in \Theta$ be irrationals sufficiently close to θ, θ' respectively such that $\text{ord}_{1,l}(\theta) = \overline{\text{ord}}_{1,l}(\theta)$ and $\text{ord}_{1,l}(\theta') = \overline{\text{ord}}_{1,l}(\theta')$. In this case, clearly the order of the points $\{1, \mathbb{x}_\theta^{[l]}(1)\}$ is different from the order of $\{1, \mathbb{x}_{\theta'}^{[l]}(1)\}$.

Case 2: $\varepsilon_n = \varepsilon'_n$ for all $n \geq 1$, and $\bar{a}_l \neq \bar{a}'_l$ for some minimal $l \geq 1$.

Without loss of generality, assume that $\bar{a}_l < \bar{a}'_l$ and let $k = \bar{a}_l$. Let θ and θ' be irrationals sufficiently close to θ and θ' such that $\text{ord}_{k+1,l}(\theta) = \overline{\text{ord}}_{k+1,l}(\theta)$ and $\text{ord}_{k+1,l}(\theta') = \overline{\text{ord}}_{k+1,l}(\theta')$. Then, the order of points $\left\{ \mathbb{x}_\theta^{q_{[l-2]}(\theta) + jq_{[l-1]}(\theta)}(1) \right\}_{0 \leq j \leq k+1}$ is different from the order of points $\left\{ \mathbb{x}_{\theta'}^{q_{[l-2]}(\theta') + jq_{[l-1]}(\theta')}(1) \right\}_{0 \leq j \leq k+1}$. $\square$

4.2. Universality of $\bar{\Theta}$. The following result is a general theorem in topological dynamics.

Theorem 4.6 (Minimal dynamical compactification). *Consider a topological dynamical system $f : X \rightarrow X$ and an embedding $h : X \rightarrow Y$ to a compact Hausdorff space Y . There exists a unique minimal compactification $(\hat{i}, \hat{X}, \hat{f})$ of (X, f) such that h admits an $\hat{i}$ -extension $\hat{h}$ in the following sense. Let $(\tilde{i}, \tilde{X}, \tilde{f})$ be any compactification of (X, f) such that h admits an $\tilde{i}$ -extension $\tilde{h}$. There exists a continuous surjective map $\phi : \tilde{X} \rightarrow \hat{X}$ such that*

$$(\dagger) \quad \phi \circ \tilde{i} = \hat{i}, \quad \phi \circ \tilde{f} = \hat{f} \circ \phi, \quad \tilde{h} = \hat{h} \circ \phi.$$

The reference embedding h is important. Without it, the theorem is false. This result is likely part of the folklore of topological dynamics. However, in the absence of an explicit reference in the standard literature, we include its proof here for the reader's convenience. The proof follows a standard topological construction via an embedding into the Hilbert cube.

Proof. Consider the compact sequence space $Y^{\mathbb{N}}$, equipped with the infinite product topology. Let us embed X into $Y^{\mathbb{N}}$ via the map

$$\Phi : X \rightarrow Y^{\mathbb{N}}, \quad \Phi(x) = \langle h \circ f^n(x) \rangle_{n \geq 0}.$$

Since h is an embedding and Y is Hausdorff, then Φ is a homeomorphism onto its image, i.e. again an embedding. It induces a conjugacy between $f : X \rightarrow X$ and the shift map on the image $\Phi(X)$. Let us set

- $\hat{X}$ to be the closure of $\Phi(X)$ as a subset of $Y^{\mathbb{N}}$,
- $\hat{i} : X \rightarrow \hat{X}$ to be equal to $\hat{i}(x) = \Phi(x)$,
- $\hat{f} : \hat{X} \rightarrow \hat{X}$ to be the shift map, and
- $\hat{h} : \hat{X} \rightarrow Y$ to be the projection to the first coordinate: $\hat{h}(\langle x_n \rangle_{n \geq 0}) = x_0$.

It is clear that $(\hat{i}, \hat{X}, \hat{f})$ is a compactification of (X, f) and $\hat{h}$ is an $\hat{i}$ -extension of h .

Next, let us prove the universality part. Pick an arbitrary compactification $(\tilde{i}, \tilde{X}, \tilde{f})$ such that h admits an $\tilde{i}$ -extension $\tilde{h}$. Consider the map

$$\phi : \tilde{X} \rightarrow \hat{X}, \quad \phi(z) = \left\langle \tilde{h} \circ \tilde{f}^n(z) \right\rangle_{n \geq 0}.$$

Observe that $\phi \circ \tilde{i} = \hat{i}$ and so the image $\phi(\tilde{i}(X))$ is dense in $\hat{X}$. Since $\tilde{i}(X)$ is dense in $\tilde{X}$ and ϕ is continuous, then the image of ϕ has to be equal to $\hat{X}$. Therefore,

ϕ induces a continuous surjective map $\phi : \tilde{X} \rightarrow \hat{X}$. It is straightforward to check that the three equations in $(\dagger)$ hold. $\square$

Theorem 4.7. *The unique minimal compactification of $(\Theta, \mathfrak{g})$ that respects the embedding $\eta : \Theta \rightarrow [0, 1]$ is homeomorphic to $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$.*

Proof. Consider the minimal compactification $(\hat{\iota}, \hat{\Theta}, \hat{\mathfrak{g}})$ of $(\Theta, \mathfrak{g})$ respecting η constructed in Theorem 4.6. Let $\hat{\eta}$ be the $\hat{\iota}$ -extension of η . From Proposition 4.4, we know that the compactification $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$ respects η with $\bar{\iota}$ -extension $\bar{\eta}$. Consider the continuous surjective map $\phi : \bar{\Theta} \rightarrow \hat{\Theta}$ from Theorem 4.6. It can be written explicitly as

$$\phi : \bar{\Theta} \rightarrow \hat{\Theta} \subset [0, 1]^{\mathbb{N}}, \quad \phi(\theta) = \langle \bar{\eta} \circ \bar{\mathfrak{g}}^n(\theta) \rangle_{n \geq 0}.$$

More explicitly, for every element $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}$ of $\bar{\Theta}$, we have $\phi(\theta) = \langle x_n \rangle_{n \geq 0}$ where for all $n \geq 1$,

$$x_{n-1} = \frac{1 - \varepsilon_n}{2} + [\varepsilon_n b_n(\theta), \varepsilon_{n+1} b_{n+1}(\theta), \dots]_-.$$

We need to show that ϕ is a homeomorphism. Since $\bar{\Theta}$ is Hausdorff and ϕ is surjective, it remains to show that ϕ is injective.

Suppose for a contradiction that ϕ is not injective. There exist two distinct elements $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1}$ and $\theta' = \langle (\varepsilon'_n, \bar{a}'_n) \rangle_{n \geq 1}$ of $\bar{\Theta}$ such that $\langle x_n \rangle_{n \geq 0} = \phi(\theta)$ is equal to $\langle x'_n \rangle_{n \geq 0} = \phi(\theta')$. After shifting, we will assume without loss of generality that $(\varepsilon_1, \bar{a}_1) \neq (\varepsilon_2, \bar{a}_2)$. There are two cases.

- Suppose $\varepsilon_1 \neq \varepsilon'_1$. Without loss of generality, suppose $\varepsilon_1 = +$ and $\varepsilon_2 = -$. Since $x_0 = x'_0$, then both x_0 and x'_0 must be equal to $\frac{1}{2}$. Therefore, $\bar{a}_1 = \bar{a}'_1 = 2$, $(\varepsilon_2, \bar{a}_2) = (-, \infty)$, and $(\varepsilon'_2, \bar{a}'_2) = (+, \infty)$. But this would imply that $x_1 = 1$ and $x'_1 = 0$, which are not equal.
- Suppose $\varepsilon_1 = \varepsilon'_1$ and $\bar{a}_1 \neq \bar{a}'_1$. Without loss of generality, suppose $\varepsilon_1 = +$ and $\bar{a}_1 < \bar{a}'_1$. Since $x_0 = x'_0$, then both x_0 and x'_0 must be equal to $\frac{1}{k}$ for some integer $k \geq 3$. Therefore, $\bar{a}_1 = k - 1$, $\bar{a}'_1 = k$, $(\varepsilon_2, \bar{a}_2) = (+, \infty)$, and $(\varepsilon_1, \bar{a}_1) = (-, \infty)$. But this would again imply that $x_1 = 0$ and $x'_1 = 1$, which are not equal. $\square$

Theorem 4.8. *The unique minimal compactification of $(\Theta, \mathfrak{g})$ that respects the dynamical embedding is homeomorphic to $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$.*

Proof. Consider the minimal compactification $(\hat{\iota}, \hat{\Theta}, \hat{\mathfrak{g}})$ of $(\Theta, \mathfrak{g})$ respecting ord constructed in Theorem 4.6. Let $\hat{\text{ord}}$ be the $\hat{\iota}$ -extension of ord . From Proposition 4.5, we know that the compactification $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$ respects ord with $\bar{\iota}$ -extension ord . Consider the continuous surjective map $\phi : \bar{\Theta} \rightarrow \hat{\Theta}$ from Theorem 4.6. It can be written explicitly as

$$\phi : \bar{\Theta} \rightarrow \hat{\Theta} \subset \Omega^{\mathbb{N}}, \quad \phi(\theta) = \langle \overline{\text{ord}} \circ \bar{\mathfrak{g}}^n(\theta) \rangle_{n \geq 0}.$$

To prove that ϕ is a homeomorphism, we just need to show that ϕ is injective. But this follows directly from the fact that $\overline{\text{ord}}$ is an embedding (Proposition 4.5). $\square$

The three theorems above imply that taking ord into account is equivalent to taking η into account.

Corollary 4.9. *Every compactification of $(\Theta, \mathfrak{g})$ that respects $\eta : \Theta \rightarrow [0, 1]$ also respects $\text{ord} : \Theta \rightarrow \Omega$, and vice versa.*

Proof. Let $(\tilde{\iota}, \tilde{\Theta}, \tilde{\mathfrak{g}})$ be a compactification such that η admits an $\tilde{\iota}$ -extension $\tilde{\eta}$. By Theorems 4.6 and 4.7, there exists a continuous surjective map $\phi : \tilde{\Theta} \rightarrow \bar{\Theta}$ such that

$$\phi \circ \tilde{\iota} = \bar{\iota}, \quad \phi \circ \tilde{\mathfrak{g}} = \bar{\mathfrak{g}} \circ \phi, \quad \tilde{\eta} = \bar{\eta} \circ \phi.$$

Let us define $\widetilde{\text{ord}} : \tilde{\Theta} \rightarrow \Omega$ by $\widetilde{\text{ord}} = \bar{\text{ord}} \circ \phi$. It satisfies

$$\widetilde{\text{ord}} \circ \tilde{\iota} = \bar{\text{ord}} \circ \phi \circ \tilde{\iota} = \bar{\text{ord}} \circ \bar{\iota} = \text{ord}.$$

Hence, $\widetilde{\text{ord}}$ is indeed an $\tilde{\iota}$ -extension of ord . The converse is analogous. $\square$

Altogether the results above imply Theorem B. In contrast to the previous two theorems, we have:

Proposition 4.10. *The unique minimal compactification of $(\Theta, \mathfrak{g})$ that respects the multiplier map $\mu : \Theta \rightarrow \mathbb{T}$ is not homeomorphic to $(\bar{\iota}, \bar{\Theta}, \bar{\mathfrak{g}})$. In particular, the map*

$$\bar{\Theta} \rightarrow \mathbb{T}^{\mathbb{N}}, \quad \theta \mapsto \langle \bar{\mu}(\mathfrak{s}^n(\theta)) \rangle_{n \geq 0}$$

is not injective.

Proof. Here is one example. Let $\theta \in \bar{\Theta}$ be of the form $\langle (\varepsilon_1, \infty), (\varepsilon_2, \infty), (\varepsilon_3, \infty), \dots \rangle$. Regardless of the value of ε_n 's, we have that for all $n \geq 0$, $\bar{\mu}(\mathfrak{s}^n(\theta)) \equiv 0 \pmod{1}$. $\square$

4.3. The time group.

Definition 4.11. For every $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \bar{\Theta}$, we define an additive abelian group $\mathcal{T}_\theta^{\text{gp}}$, called the *time group of θ* , such that it is generated by the sequence $\{\mathfrak{q}_{[n]}\}_{n \geq 0}$ subject to the relations

$$\mathfrak{q}_{[n+1]} = b_{n+1}\mathfrak{q}_{[n]} - \varepsilon_n \varepsilon_{n+1} \mathfrak{q}_{[n-1]}, \quad \text{where } b_{n+1} = \bar{a}_{n+1} + \frac{1 + \varepsilon_{n+1} \varepsilon_{n+2}}{2},$$

whenever n is not a parabolic level, and where $\mathfrak{q}_{[-1]} = 0$ is the identity element. For $n \geq 0$, the element $\mathfrak{q}_{[n]} = \mathfrak{q}_{[n]}(\theta) \in \mathcal{T}_\theta^{\text{gp}}$ will be referred to as the n^{th} *return time of θ* .

Proposition 4.12 (Chronological order). *For every $\theta = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \geq 1} \in \bar{\Theta}$, the time group $\mathcal{T}_\theta^{\text{gp}}$ admits a unique translation-invariant total order $<$, called the chronological order of $\mathcal{T}_\theta$, with the property that for all $n \geq 0$,*

- (1) $0 < \mathfrak{q}_{[n]} < \mathfrak{q}_{[n+1]}$,
- (2) if $\bar{a}_{n+1} = \infty$, then $k\mathfrak{q}_{[n]} < \mathfrak{q}_{[n+1]}$ for all $k \geq 1$.

The positive cone of $<$ is the commutative semigroup

$$\mathcal{T}_\theta := \{P \in \mathcal{T}_\theta^{\text{gp}} : P > 0\},$$

which we refer to as the *time semigroup of θ* .

Proof. The order $<$ is uniquely characterized by the semigroup $\mathcal{T}_\theta$ since $a < b$ if and only if $b - a \in \mathcal{T}_\theta$. Hence, it is sufficient to construct the time semigroup $\mathcal{T}_\theta$ and claim that $\mathcal{T}_\theta^{\text{gp}}$ is the disjoint union of $\mathcal{T}_\theta$, $\{0\}$, and $-\mathcal{T}_\theta$.

Let $\mathcal{T}_\theta$ be the semigroup generated by

$$\{\mathfrak{q}_{[0]}\} \cup \{\mathfrak{q}_{[n+1]} - \mathfrak{q}_{[n]}\}_{n \geq 0} \cup \bigcup_{\bar{a}_{n+1} = \infty} \{\mathfrak{q}_{[n+1]} - k\mathfrak{q}_{[n]}\}_{k \geq 2}.$$

Let us prove by induction over $m \geq 0$ the following statement:

$(\heartsuit_m)$ For every $(c_0, c_1, \dots, c_m) \in \mathbb{Z}^{m+1}$, the sum $\sum_{j=0}^m c_j \mathfrak{q}_{[j]}$ is contained in exactly one of the following sets: $\mathcal{T}_\theta$, $\{0\}$, $-\mathcal{T}_\theta$.

Since $\mathfrak{q}_{[0]}$ is in $\mathcal{T}_\theta$, $(\heartsuit_0)$ holds. Suppose $(\heartsuit_k)$ holds for some $k \geq 0$. Pick any sequence of integers $c_0, c_1, \dots, c_{k+1}$ with $c_{k+1} \neq 0$ and let $x = \sum_{j=0}^{k+1} c_j \mathfrak{q}_{[j]}$. There are two cases.

- Suppose $\bar{a}_{k+1} < \infty$. Since $\mathfrak{q}_{[k+1]}$ is an integral combination of $\mathfrak{q}_{[k]}$ and $\mathfrak{q}_{[k-1]}$, then by $(\heartsuit_k)$, we have $x \in \mathcal{T}_\theta \cup \{0\} \cup -\mathcal{T}_\theta$.
- Suppose $\bar{a}_{k+1} = \infty$. Since $\mathfrak{q}_{[k+1]}$ is in $\mathcal{T}_\theta$, it suffices to show that if $c_{k+1} = 1$, then $x \in \mathcal{T}_\theta$. Hence, we will now assume that $c_{k+1} = 1$. For $j \in \{0, \dots, k\}$, denote $d_j = \sum_{j=0}^k \max\{0, -c_j\}$, which satisfies $d_j \geq 0$ and $c_j + d_j \geq 0$. Let $d = \sum_{j=0}^k d_j \geq 0$. Then,

$$\begin{aligned} x &= \sum_{j=0}^k c_j \mathfrak{q}_{[j]} + \mathfrak{q}_{[k+1]} \\ &= \sum_{j=0}^k (c_j + d_j) \mathfrak{q}_{[j]} + \sum_{j=0}^k d_j (\mathfrak{q}_{[k]} - \mathfrak{q}_{[j]}) + (\mathfrak{q}_{[k+1]} - d \mathfrak{q}_{[k]}). \end{aligned}$$

The last term in the summation above is contained in $\mathcal{T}_\theta$, and each of the remaining terms is contained in $\mathcal{T}_\theta \cup \{0\}$. Therefore, x is in $\mathcal{T}_\theta$.

Hence, by induction, $(\heartsuit_k)$ holds for all k , and we are done. $\square$

If θ is irrational, then $\mathfrak{q}_{[0]} \mapsto 1$ induces an isomorphism between $(\mathcal{T}_\theta^{\text{gp}}, <)$ and the ordered group of integers $(\mathbb{Z}, <)$ that sends each $\mathfrak{q}_{[n]}$ to the number $q_{[n]} \in \mathbb{N}$ coming from §2.2. Otherwise, $(\mathcal{T}_\theta^{\text{gp}}, <)$ is a non-Archimedean ordered group.

In practice, sometimes it is more convenient to consider a different set of generators $\{q_n = \mathfrak{q}_n(\theta)\}_{n \geq 0}$ for $\mathcal{T}_\theta^{\text{gp}}$. It mimics the denominator continuants of the regular continued fraction expansion and it is defined together with an increasing sequence of integers $\{t_n\}_{n \geq 0}$ and a sequence $\{a_n\}_{n \geq 1}$ of elements of $\bar{\mathbb{N}}$ as follows. First, set

$$t_0 = 0, \quad a_1 = \bar{a}_1, \quad \text{and} \quad \mathfrak{q}_0 = \mathfrak{q}_{[0]}.$$

The rest is defined inductively as follows. For $n \geq 0$,

- if $\varepsilon_{n+1}\varepsilon_{n+2} = -1$, then

$$t_{n+1} = t_n + 1, \quad a_{t_{n+2}} = \bar{a}_{n+2}, \quad \text{and} \quad \mathfrak{q}_{t_{n+1}} = \mathfrak{q}_{[n+1]};$$

- if $\varepsilon_{n+1}\varepsilon_{n+2} = +1$, then

$$\begin{aligned} t_{n+1} &= t_n + 2, & a_{t_{n+2}} &= 1, & a_{t_{n+3}} &= \bar{a}_{n+2} - 1, \\ \mathfrak{q}_{t_{n+1}} &= \mathfrak{q}_{[n+1]} - \mathfrak{q}_{[n]}, & \text{and} & & \mathfrak{q}_{t_{n+1}} &= \mathfrak{q}_{[n+1]}. \end{aligned}$$

We will call $q_n = \mathfrak{q}_n(\theta)$ the n^{th} slow return time of θ . The group relations become slightly simpler, namely

$$q_{n+1} = a_{n+1}q_n + q_{n-1} \quad \text{for } n \geq 0 \text{ with } a_{n+1} < \infty,$$

where $\mathbf{q}_{-1} = 0$ is set to be the identity for convenience. In this format, the set of generators of the time semigroup $\mathcal{T}_\theta$ is

$$\{\mathbf{q}_n\}_{n \geq 0} \cup \bigcup_{a_{n+1} = \infty} \{\mathbf{q}_{n+1} - k\mathbf{q}_n\}_{k \geq 1}.$$

5. THE NATURAL EXTENSION

Consider the bi-infinite sequence space

$$\overline{\underline{\Theta}} := \overline{\Sigma}^{\mathbb{Z}} = \{ \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} : (\varepsilon_n, \bar{a}_n) \in \overline{\Sigma} \text{ for all } n \},$$

equipped with the infinite product topology. The projection map

$$\text{proj} : \overline{\underline{\Theta}} \rightarrow \overline{\Theta}, \quad \langle \dots, (\varepsilon_{-1}, \bar{a}_{-1}), (\varepsilon_0, \bar{a}_0); (\varepsilon_1, \bar{a}_1), \dots \rangle \mapsto \langle (\varepsilon_1, \bar{a}_1), (\varepsilon_2, \bar{a}_2), \dots \rangle$$

is continuous and the shift map $\mathfrak{s} : \overline{\underline{\Theta}} \rightarrow \overline{\underline{\Theta}}$ given by

$$\mathfrak{s}(\langle \dots, (\varepsilon_{-1}, \bar{a}_{-1}), (\varepsilon_0, \bar{a}_0); (\varepsilon_1, \bar{a}_1), \dots \rangle) = \langle \dots, (\varepsilon_0, \bar{a}_0), (\varepsilon_1, \bar{a}_1); (\varepsilon_2, \bar{a}_2), \dots \rangle$$

is a homeomorphism. Indeed, $\mathfrak{s} : \overline{\underline{\Theta}} \rightarrow \overline{\underline{\Theta}}$ is the natural extension of $\mathfrak{s} : \overline{\Theta} \rightarrow \overline{\Theta}$.

5.1. The time group.

Definition 5.1. For every $\underline{\theta} = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} \in \overline{\underline{\Theta}}$, we again define an additive abelian group $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$, called the *time group* of $\underline{\theta}$, such that it is generated by the sequence $\{Q_{[n]}\}_{n \in \mathbb{Z}}$ and subject to the relations

$$Q_{[n+1]} = b_{n+1}Q_{[n]} - \varepsilon_n \varepsilon_{n+1} Q_{[n-1]} \quad \text{where} \quad b_{n+1} = \bar{a}_{n+1} + \frac{1 + \varepsilon_{n+1} \varepsilon_{n+2}}{2}$$

for all $n \in \mathbb{Z}$ with $\bar{a}_{n+1} < \infty$.

For every $n \in \mathbb{Z}$, $Q_{[n]} = Q_{[n]}(\underline{\theta})$ is called the n^{th} return time of $\underline{\theta}$. Below, we will again define another generating set $\{Q_n = Q_n(\underline{\theta})\}_{n \in \mathbb{Z}}$ for $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$. To do so, we will need a strictly increasing sequence of integers $\{t_n\}_{n \in \mathbb{Z}}$ and a sequence of positive integers $\{a_n\}_{n \in \mathbb{Z}}$. First, we set $t_0 = 0$. For all $n \geq 1$, t_n and t_{-n} are defined recursively by

$$t_n = \begin{cases} t_{n-1} + 1 & \text{if } \varepsilon_n \varepsilon_{n+1} = -, \\ t_{n-1} + 2 & \text{if } \varepsilon_n \varepsilon_{n+1} = +, \end{cases}$$

and

$$t_{-n} = \begin{cases} t_{-n+1} - 1 & \text{if } \varepsilon_{-n+1} \varepsilon_{-n+2} = -, \\ t_{-n+1} - 2 & \text{if } \varepsilon_{-n+1} \varepsilon_{-n+2} = +. \end{cases}$$

For all $n \in \mathbb{Z}$,

- if $\varepsilon_n \varepsilon_{n+1} = -$, set $a_{t_{n+1}} = \bar{a}_{n+1}$;
- if $\varepsilon_n \varepsilon_{n+1} = +$, set $a_{t_n} = 1$ and $a_{t_{n+1}} = \bar{a}_{n+1} - 1$.

For all $n \in \mathbb{Z}$, we set

$$Q_{t_n} := Q_{[n]},$$

and whenever $\varepsilon_n \varepsilon_{n+1} = +$, we set

$$Q_{t_{n-1}} := Q_{[n]} - Q_{[n-1]}.$$

The corresponding set of relations for this new generating set is simpler: for every $m \in \mathbb{Z}$ with $a_m < \infty$, we have

$$Q_m = a_m Q_{m-1} + Q_{m-2}.$$

Proposition 5.2 (Chronological order). *For every $\underline{\theta} = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} \in \bar{\Theta}$, the time group $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$ admits a unique translation-invariant total order $<$, called the chronological order of $\mathbf{T}_{\underline{\theta}}$, with the property that for all $n \in \mathbb{Z}$,*

- (1) $0 < Q_{[n]} < Q_{[n+1]}$,
- (2) if $\bar{a}_{n+1} = \infty$, then $kQ_{[n]} < Q_{[n+1]}$ for all $k \geq 1$.

The positive cone of $<$ is the commutative semigroup

$$\mathbf{T}_{\underline{\theta}} := \{P \in \mathbf{T}_{\underline{\theta}}^{\text{gp}} : P > 0\},$$

which we refer to as the *time semigroup* of $\underline{\theta}$.

Proof. The idea is similar to the proof of Proposition 4.12. Consider the slow generating set $\{Q_n\}_{n \in \mathbb{Z}}$. We define $\mathbf{T}_{\underline{\theta}}$ to be the semigroup generated by

$$\{Q_n\}_{n \in \mathbb{Z}} \cup \bigcup_{a_{n+1} = \infty} \{Q_{n+1} - kQ_n\}_{n \in \mathbb{Z}}.$$

We need to show that $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$ is the disjoint union of $\mathbf{T}_{\underline{\theta}}$, $\{0\}$, and $-\mathbf{T}_{\underline{\theta}}$.

Let us prove by induction over $m \geq 0$ the following statement:

- ($\diamond_m$) For every $(n, c_0, c_1, \dots, c_m) \in \mathbb{Z}^{m+2}$, the sum $\sum_{j=0}^m c_j Q_{n+m}$ is contained in exactly one of the following sets: $\mathcal{T}_{\underline{\theta}}$, $\{0\}$, $-\mathcal{T}_{\underline{\theta}}$.

Similar to the proof of Proposition 4.12, ($\diamond_0$) is trivial and for $m \geq 1$, ($\diamond_m$) implies ($\diamond_{m+1}$). It remains to prove ($\diamond_1$).

Let us pick $n \in \mathbb{Z}$ and suppose x is an integral combination of Q_n and Q_{n-1} . If the coefficients of Q_n and Q_{n-1} are either zero or have the same sign, then clearly x will belong in either $\mathcal{T}_{\underline{\theta}}$, or $\{0\}$, or $-\mathcal{T}_{\underline{\theta}}$. We can now assume that

$$x = c_0 Q_n - d_0 Q_{n-1}$$

for some co-prime positive integers c_0 and d_0 . We will assume that $0 < c_0 < d_0$ because otherwise x can be expressed as the sum of $(c_0 - d_0)Q_n \in \mathbf{T}_{\underline{\theta}} \cup \{0\}$ and $d_0(Q_n - Q_{n-1}) \in \mathbf{T}_{\underline{\theta}}$ and we are done. To proceed, there are two cases.

- (i) Suppose $a_n = \infty$. We can present x as

$$x = (c_0 - 1)Q_n + (Q_n - d_0 Q_{n-1}).$$

In the expression above, the first term is in $\mathbf{T}_{\underline{\theta}} \cup \{0\}$ and the second term is in $\mathbf{T}_{\underline{\theta}}$. Hence, x is in $\mathbf{T}_{\underline{\theta}}$.

- (ii) Suppose $a_n < \infty$, so then we can write

$$x = -(c_{-1}Q_{n-1} - d_{-1}Q_{n-2}) \quad \text{where } c_{-1} = d_0 - c_0 a_n \text{ and } d_{-1} = c_0.$$

There are three sub-cases.

- (a) Suppose $c_{-1} \leq 0$. Since x is the sum of $(-c_{-1})Q_{n-1} \in \mathbf{T}_{\underline{\theta}} \cup \{0\}$ and $d_{-1}Q_{n-2} \in \mathbf{T}_{\underline{\theta}}$, then x is in $\mathbf{T}_{\underline{\theta}}$.
- (b) Suppose $c_{-1} \geq d_{-1}$. Since x is the sum of $-(c_{-1} - d_{-1})Q_{n-1} \in -\mathbf{T}_{\underline{\theta}} \cup \{0\}$ and $-d_{-1}(Q_{n-1} - Q_{n-2}) \in -\mathbf{T}_{\underline{\theta}}$, then x is in $-\mathbf{T}_{\underline{\theta}}$.
- (c) Suppose $0 < c_{-1} < d_{-1}$. This assumption is equivalent to

$$\frac{1}{a_n + 1} < \frac{c_0}{d_0} < \frac{1}{a_n}.$$

Analogous to (i), we have $x \in -\mathbf{T}_{\underline{\theta}}$ if $a_{n-1} = \infty$. Suppose $a_{n-1} < \infty$. Then, we can write

$$x = c_{-2}Q_{n-2} - d_{-2}Q_{n-3}$$

where

$$c_{-2} = d_{-1} - c_{-1}a_{n-1} \quad \text{and} \quad d_{-2} = c_{-1}.$$

Again, analogous to (2) (a)–(b), we are done if $c_{-2} \notin (0, d_{-2})$ and this leaves us with the case where $0 < c_{-2} < d_{-2}$. This amounts to

$$\frac{1}{a_n + \frac{1}{a_{n-1}}} < \frac{c_0}{d_0} < \frac{1}{a_n + \frac{1}{a_{n-1}+1}}.$$

If we end up having to repeat the argument above $k \geq 1$ times, we are left with the case where $a_n, a_{n-1}, \dots, a_{n-k+1} < \infty$ and x can be written as

$$x = (-1)^k (c_{-k}Q_{n-k} - d_{-k}Q_{n-k-1})$$

where

$$\begin{bmatrix} c_{-k} \\ d_{-k} \end{bmatrix} = \begin{bmatrix} -a_{n-k+1} & 1 \\ 1 & 0 \end{bmatrix} \cdots \begin{bmatrix} -a_n & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c_0 \\ d_0 \end{bmatrix}.$$

In this case, $\frac{c_0}{d_0}$ is between $[a_n, a_{n-1}, \dots, a_{n-k+1}]$ and $[a_n, a_{n-1}, \dots, a_{n-k+1} + 1]$. Eventually, this process has to stop at some value k because otherwise the rational number $\frac{c_0}{d_0}$ would be arbitrarily close to the irrational $[a_n, a_{n-1}, a_{n-2}, \dots]$. We are done with the proof of $(\diamond_1)$. $\square$

Next, we will define another total order $\triangleright$ on $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$.

Proposition 5.3 (Left-right order). *For $\underline{\theta} \in \overline{\Theta}$, the time group $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$ admits a unique translation invariant total order $\triangleleft$, called the left-right order, that satisfies the following properties.*

- (1) *If $\varepsilon_1 = +1$, then $Q_0 \triangleleft 0$; otherwise, $\varepsilon_1 = -1$ and $0 \triangleleft Q_0$.*
- (2) *For $n \in \mathbb{Z}$, $-Q_n$ is always $\triangleleft$ -between 0 and Q_{n-1} .*
- (3) *For $n \in \mathbb{Z}$ with $a_{n+1} = \infty$, then $-kQ_n$ is always $\triangleleft$ -between 0 and Q_{n-1} for all $k \geq 1$.*

We will call $\triangleright$ the *left-right order* on $\overline{\Theta}$. It naturally descends to a translation-invariant total order $\triangleright$ on the time semigroup $\mathbf{T}_{\underline{\theta}}$. The geometric meaning behind $\triangleleft$ will become apparent in the next section.

Proof. Assume $\varepsilon_1 = -$ and we will have $0 \triangleleft Q_0$. Let $Q'_n = (-1)^n Q_n$. It satisfies the relation $Q'_{n-1} = a_{n+1}Q'_n + Q'_{n+1}$ for all $n \in \mathbb{Z}$ with $a_{n+1} < \infty$. Let Υ be the semigroup generated by

$$\{Q'_n\}_{n \in \mathbb{Z}} \cup \bigcup_{a_{n+1} = \infty} \{Q'_{n-1} - kQ'_n\}_{k \geq 1}.$$

With almost the same proof as in Proposition 5.2, the time group $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$ is indeed equal to the disjoint union of Υ , $\{0\}$, and $-\Upsilon$. This induces a total order $\triangleleft$ on $\mathbf{T}_{\underline{\theta}}^{\text{gp}}$ such that Υ is the corresponding positive cone, that is, $P \triangleleft Q$ if and only if $Q - P \in \Upsilon$.

It remains to show that $\triangleleft$ satisfies the desired properties. Property (1) is clear. By the construction, we have that for every odd integer n ,

$$-Q_n \triangleleft 0 \triangleleft Q_{n+1} \triangleleft Q_{n-1},$$

which implies property (2). For all n with $a_{n+1} = \infty$ and all $k \geq 1$, since both Q'_n and $Q'_{n-1} - kQ'_n$ are in Υ , then $0 \triangleleft kQ'_n \triangleleft Q'_{n-1}$ and this implies property (3).

Lastly, if $\varepsilon_1 = +1$, then we will take Υ as the negative cone of $\triangleleft$. The rest of the proof above is exactly the same. $\square$

The dynamical meaning of $\triangleleft$ will be clarified in Proposition 5.7.

5.2. Cascade of translations. Consider the subspace

$$\underline{\Theta} := \Sigma^{\mathbb{Z}} = \left\{ \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} \in \bar{\Theta} : \bar{a}_n < \infty \text{ for all } n \right\}$$

of $\bar{\Theta}$. We have the following commutative diagram.

$$\begin{array}{ccc} \mathfrak{s} \hookrightarrow \underline{\Theta} & \xhookrightarrow{\quad} & \bar{\Theta} \hookrightarrow \mathfrak{s} \\ \mathfrak{X} \circ \text{proj} \downarrow & & \downarrow \text{proj} \\ \mathfrak{g} \hookrightarrow \Theta & \xrightarrow{\quad \bar{i} \quad} & \bar{\Theta} \hookrightarrow \mathfrak{s} \end{array}$$

Every element $\underline{\theta}$ of $\underline{\Theta}$ induces a bi-infinite sequence of irrationals

$$\theta_n = \mathfrak{X} \circ \text{proj} \circ \mathfrak{s}^n(\underline{\theta}) \in \Theta, \quad n \in \mathbb{Z},$$

where $\mathfrak{g}(\theta_n) = \theta_{n+1}$ for all n . The bi-infinite tower of sector renormalizations

$$\{\mathfrak{x}_{\theta_n} : \bar{\mathbb{D}} \rightarrow \bar{\mathbb{D}}\}_{n \in \mathbb{Z}}$$

can be put into a single dynamical plane as follows.

Definition 5.4. For $\underline{\theta} = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} \in \underline{\Theta}$, we define the (full) cascade associated to $\underline{\theta}$ to be the abelian group $F = F_{\underline{\theta}}$ of translations generated by $\{F^{[n]} = F_{\underline{\theta}}^{[n]}\}_{n \in \mathbb{Z}}$ where for $n \in \mathbb{Z}$,

$$F^{[n]}(z) := z - \varepsilon_{n+1} l_{[n]} \quad \text{where } l_{[n]} := l_{[n]}(\underline{\theta}) := \begin{cases} |\theta_0 \dots \theta_n| & \text{if } n \geq 0, \\ 1 & \text{if } n = -1, \\ |\theta_{-1} \dots \theta_{-n+1}|^{-1} & \text{if } n \leq -2, \end{cases}$$

Let us fix $\underline{\theta} = \langle (\varepsilon_n, \bar{a}_n) \rangle_{n \in \mathbb{Z}} \in \underline{\Theta}$. For $n \in \mathbb{Z}$, denote $b_n = \bar{a}_n + \frac{1 + \varepsilon_n \varepsilon_{n+1}}{2}$.

Lemma 5.5. For all $n \in \mathbb{Z}$, we have

$$F^{[n]} = (F^{[n-1]})^{b_n} \circ (F^{[n-2]})^{-\varepsilon_{n-1} \varepsilon_n}.$$

Proof. From Lemma 2.1, we have $\theta_n = \varepsilon_n b_n - \frac{1}{\theta_{n-1}}$. Therefore,

$$\varepsilon_{n+1} |\theta_{n-1} \theta_n| = |\theta_{n-1}| \left(\varepsilon_n b_n - \frac{1}{\theta_{n-1}} \right) = b_n \cdot \varepsilon_n |\theta_{n-1}| - \varepsilon_{n-1} \varepsilon_n \cdot \varepsilon_{n-1}.$$

By multiplying both sides by $l_{[n-2]}$, we obtain

$$\varepsilon_{n+1} l_{[n]} = b_n l_{[n-1]} - \varepsilon_{n-1} \varepsilon_n l_{[n-2]}.$$

This implies the desired equation. $\square$

Consider the time group $\mathbf{T}^{\text{sp}} = \mathbf{T}_{\underline{\theta}}^{\text{sp}}$ of $\underline{\theta}$ defined in the previous section.

Proposition 5.6. The time group $\mathbf{T}^{\text{sp}}$ of $\underline{\theta}$ is isomorphic to the full cascade F associated to $\underline{\theta}$ via the identification $Q_{[n]} \mapsto F^{[n]}$ for all $n \in \mathbb{Z}$.

Proof. According to the previous lemma, there is a well-defined surjective homomorphism

$$\phi : \mathbf{T}^{\text{gp}} \rightarrow F, \quad P \mapsto F^P,$$

where $F^{Q_{[n]}} = F^{[n]}$ for every $n \in \mathbb{Z}$. It remains to check that ϕ is injective. Pick $P \in \mathbf{T}^{\text{gp}}$, so P can be written as an integral combination of $Q_{[n]}, Q_{[n+1]}, \dots, Q_{[n+k]}$ for some integers $n \in \mathbb{Z}$ and $k \geq 0$. By the recurrence relation, we can rewrite P as $c_0 Q_{[n]} + c_1 Q_{[n+1]}$ for some integer coefficients c_0, c_1 . Hence,

$$F^P(z) = z - \varepsilon_{n+1} c_0 l_{[n]} - \varepsilon_{n+2} c_1 l_{[n+1]} = z - (\varepsilon_{n+1} c_0 + \varepsilon_{n+1} c_1 |\theta_{n+1}|) l_{[n]}.$$

Since θ_{n+1} is irrational, the equation $\varepsilon_{n+1} c_0 + \varepsilon_{n+1} c_1 |\theta_{n+1}| = 0$ only has trivial solution $c_0 = c_1 = 0$. This implies that F^P is the identity map if and only if $P = 0$. Therefore, ϕ is injective. $\square$

As a consequence of this proposition, we can parametrize F in terms of elements of $\mathbf{T}^{\text{gp}}$ and write

$$F = (F^P)_{P \in \mathbf{T}^{\text{gp}}} \quad \text{where} \quad F^{Q_{[n]}} \equiv F^{[n]}.$$

From now on, we will denote

$$V_P := F^P(0) \quad \text{for } P \in \mathbf{T}^{\text{gp}}.$$

By Propositions 4.12 and 5.2, F admits two natural total orders, namely the chronological order $<$ and the left-right order $\triangleleft$. The latter is called the left-right order for the following reason.

Proposition 5.7. *For any two elements P_1 and P_2 in $\mathbf{T}^{\text{gp}}$,*

$$V_{P_1} < V_{P_2} \quad \text{if and only if} \quad P_1 \triangleleft P_2.$$

Proof. By Proposition 5.6, there does exist a well-defined strict total order $\triangleleft$ such that $P_1 \triangleleft P_2$ if and only if $V_{P_1} < V_{P_2}$. It is clear that $\triangleleft$ is translation-invariant since F simply acts as translations on the real line. To prove the proposition, we just need to check that $\triangleleft$ satisfies properties (1) and (2) in Proposition 5.3. Property (3) is inapplicable because $a_n < \infty$ for all n in our current setting.

Property (1) is clear: since $V_{Q_0} = -\varepsilon_1 l_{[0]}$, then $V_{Q_0} < 0$ if $\varepsilon_1 = +1$ and $V_{Q_0} > 0$ if $\varepsilon_1 = -1$. Let us check property (2) for $n = 0$; the proof for every other n 's will be analogous. There are two cases.

- Suppose $\varepsilon_0 \varepsilon_1 = -1$. Since $V_{Q_0} = -\varepsilon_1 l_{[0]}$ and $V_{Q_{-1}} = -\varepsilon_0 = \varepsilon_1$, then $0 < \varepsilon_1 V_{-Q_0} < \varepsilon_1 V_{Q_{-1}}$. This implies that $-Q_0$ is $\triangleleft$ -between 0 and Q_{-1} .
- Suppose $\varepsilon_0 \varepsilon_1 = +1$. Then,

$$V_{Q_{-2}} = V_{Q_{[-1]}} = -\varepsilon_1 \quad \text{and} \quad V_{Q_{-1}} = V_{Q_{[0]} - Q_{[-1]}} = \varepsilon_1(1 - l_{[0]}).$$

Since $0 < l_{[0]} < \frac{1}{2}$, then $0 < \varepsilon_1 V_{-Q_{-2}} < \varepsilon_1 V_{Q_{-1}} < \varepsilon_1 V_{-Q_0}$. This implies that $-Q_{-1}$ is $\triangleleft$ -between 0 and Q_0 , and $-Q_{-1}$ is $\triangleleft$ -between 0 and Q_{-2} . $\square$

5.3. Renormalization triangulation. For any two points x and y on the real line, we will denote by $[x, y]$ the closed real interval with endpoints x and y . For $n \in \mathbb{Z}$, denote

$$J_n = [V_{-Q_{[n]}}, V_{-Q_{[n]} + \varepsilon_n \varepsilon_{n+1} Q_{[n-1]}}].$$

Lemma 5.8. *For all $n \in \mathbb{Z}$,*

- (1) $[0, V_{-2Q_{[n]} + \varepsilon_n \varepsilon_{n+1} Q_{[n-1]}}]$ is contained in the interior of J_n ;
- (2) J_n is contained in the interior of J_{n-1} .

Proof. We have $V_{-Q_{[n]}} = \varepsilon_{n+1}l_{[n]}$ and $V_{-Q_{[n]}+\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}} = \varepsilon_{n+1}(l_{[n]} - l_{[n-1]})$ and $V_{-2Q_{[n]}+\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}} = \varepsilon_{n+1}(2l_{[n]} - l_{[n-1]})$. Then, item (1) follows from the inequality $l_{[n-1]} > 2l_{[n]}$. Item (2) follows from:

$$\max_{x \in \partial J_n} |x| = l_{[n-1]} - l_{[n]} < l_{[n-1]} = \min_{x \in \partial J_{n-1}} |x|. \quad \square$$

The first part of the lemma above tells us that J_n can be partitioned in two ways. Firstly, we have $J_n = J_{n,0} \cup J_{n,1}$ where

$$J_{n,0} = [V_{-Q_{[n]}+\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}}, 0] \quad \text{and} \quad J_{n,1} = [0, V_{-Q_{[n]}}],$$

Secondly, we have $J_n = J'_{n,0} \cup J'_{n,1}$ where

$$J'_{n,j} := F^{-Q_{[n]}+j\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}}(J_{n,j}), \quad j \in \{0, 1\}.$$

The two partitions are related by the commuting pair

$$\mathcal{F}_n = (F^{Q_{[n]}} : J'_{n,0} \rightarrow J_{n,0}, \quad F^{Q_{[n]}-\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}} : J'_{n,1} \rightarrow J_{n,1}).$$

Let us outline how the cascade F is related to $\{\mathfrak{r}_{\theta_n} : \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}\}_{n \in \mathbb{Z}}$. Denote by $-\overline{\mathbb{H}}$ the closed lower half plane. For any closed interval $J \subset \mathbb{R}$, denote the half strip

$$W(J) := \{z \in -\overline{\mathbb{H}} : \operatorname{Re} z \in J\}.$$

For $n \in \mathbb{Z}$ and $j \in \{0, 1\}$, denote

$$W_n = W(J_n), \quad W_{n,j} = W(J_{n,j}), \quad W'_{n,j} = W(J'_{n,j}).$$

Each W_n is a strip of horizontal width $l_{[n-1]}$. The second part of the lemma above implies that we have a bi-infinite nest of half-strips

$$\dots \subset W_2 \subset W_1 \subset W_0 \subset W_{-1} \subset \dots$$

whose union is $-\overline{\mathbb{H}}$ and whose nester intersection is the negative imaginary axis. The commuting pair $\mathcal{F}_n$ mentioned before extends to a commuting pair on W_n :

$$\mathcal{F}_n = (F^{Q_{[n]}} : W'_{n,0} \rightarrow W_{n,0}, \quad F^{Q_{[n]}-\varepsilon_n\varepsilon_{n+1}Q_{[n-1]}} : W'_{n,1} \rightarrow W_{n,1}).$$

The next proposition draws the bridge between cascades and sector renormalization. The proof is elementary calculation.

Proposition 5.9. *For $n \in \mathbb{Z}$, the map*

$$\phi_n : W_n \rightarrow \overline{\mathbb{D}}, \quad \phi_n(z) = e^{-2\pi iz/l_{[n-1]}}$$

satisfies the following properties.

- (1) ϕ_n conformally sends the interior of W_n to the open unit disk $\mathbb{D}$ minus the radial slit $\{\arg z = -2\pi\theta_n\}$, and sends each of the horizontal sides of W_n to this slit.
- (2) ϕ_n projects the commuting pair $\mathcal{F}_n$ to the rotation $\mathfrak{r}_{\theta_n} : \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$, and
- (3) For $k \geq n+1$, ϕ_n maps W_k conformally onto the sector $S_{\theta_n}^{k-n}$ and the commuting pair $\mathcal{F}_k$ projects to the first return map of $\mathfrak{r}_{\theta_n}$ back to the sector $S_{\theta_n}^{k-n}$.

In particular, the proposition above says that $\mathcal{F}_n$ is precisely the first return map of F back to W_n . This gives us a triangulation of the lower half plane.

Proposition 5.10 (Renormalization triangulation). *For $n \in \mathbb{Z}$,*

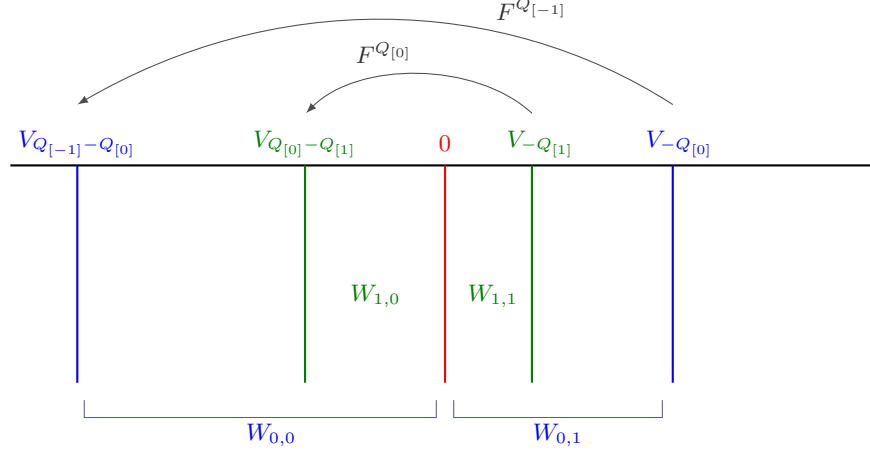


FIGURE 5. Cascade $F = F_{\underline{\theta}}$ with golden mean combinatorics $\underline{\theta} = \langle \dots, (+, 2); (+, 2), (+, 2), \dots \rangle$

- (1) *the collection of closed half-strips*

$$\mathcal{W}_n = \{F^{-P}(W_{n,0})\}_{0 \leq P < Q_{[n]}} \cup \{F^{-P}(W_{n,1})\}_{0 \leq P < Q_{[n]} - \varepsilon_n \varepsilon_{n+1} Q_{[n-1]}}$$

partitions the closed lower half plane, i.e. the elements of $\mathcal{W}_n$ have pairwise disjoint interior and their union is $\mathbb{R}$;

- (2) $\mathcal{W}_n$ is a refinement of $\mathcal{W}_{n-1}$: *every element in $\mathcal{W}_{n-1}$ is a union of at least two consecutive half-strips in $\mathcal{W}_n$.*

As a consequence, we also have:

Corollary 5.11 (Proper discontinuity). *For $n \in \mathbb{Z}$,*

$$\text{if } 0 < P < Q_{[n]} \text{ or } -Q_{[n]} < P < 0, \quad \text{then } |V_P| > l_{[n]}.$$

In particular, if P is small, then $|V_P|$ is large.

REFERENCES

- [AC17] Artur Avila and Davoud Cheraghi. Statistical properties of quadratic polynomials with a neutral fixed point. *J. Eur. Math. Soc.*, 20(8):2005–2062, 2017.
- [BC12] Xavier Buff and Arnaud Chéritat. Quadratic Julia sets with positive area. *Ann. Math.*, 176:673–746, 2012.
- [CC15] Davoud Cheraghi and Arnaud Chéritat. A proof of the Marmi-Moussa-Yoccoz conjecture for rotation numbers of high type. *Invent. Math.*, 202:677–742, 2015.
- [Che13] Davoud Cheraghi. Typical orbits of quadratic polynomials with a neutral fixed point: Brjuno type. *Commun. Math. Phys.*, 332(3):999–1035, 2013.
- [Che19] Davoud Cheraghi. Typical orbits of quadratic polynomials with a neutral fixed point: Non-Brjuno type. *Ann. Sci. Éc. Norm. Sup.*, 52(1):59–138, 2019.
- [Che25] Davoud Cheraghi. Topology of irrationally indifferent attractors. *Ann. Sci. Éc. Norm. Sup.*, 6(58):1321–1383, 2025.
- [dF99] Edson de Faria. Asymptotic rigidity of scaling ratios for critical circle mappings. *Ergod. Theory Dyn. Syst.*, 19(4):995–1035, 1999.
- [dFdM99] Edson de Faria and Wellington de Melo. Rigidity of critical circle mappings II. *J. Amer. Math. Soc.*, 13:343–370, 1999.
- [DL22] Dzmitry Dudko and Mikhail Lyubich. Uniform a priori bounds for neutral renormalization, 2022. arXiv.2210.09280.

- [DL23] Dzmitry Dudko and Mikhail Lyubich. Local connectivity of the Mandelbrot set at some satellite parameters of bounded type. *Geom. Funct. Anal.*, 33(4):912–1047, 2023.
- [DL26] Dzmitry Dudko and Mikhail Lyubich. Uniform a priori bounds for neutral renormalization. variation i: Sector renormalization, 2026. Manuscript.
- [DLL26] Dzmitry Dudko, Willie Rush Lim, and Mikhail Lyubich. Rigidity of the attractor of neutral quadratic polynomials, 2026. Manuscript.
- [DLS20] Dzmitry Dudko, Mikhail Lyubich, and Nikita Selinger. Pacman renormalization and self-similarity of the Mandelbrot set near Siegel parameters. *J. Amer. Math. Soc.*, 33(3):653–733, 2020.
- [Dou87] Adrien Douady. Disques de Siegel et anneaux de Herman. In *Séminaire Bourbaki : volume 1986/87, exposés 669-685*, number 152-153 in Astérisque, pages 4, 151–172. Société mathématique de France, 1987.
- [Ghy84] Étienne Ghys. Transformations holomorphes au voisinage d’une courbe de Jordan. *C. R. Acad. Sci. Paris Sér. I Math.*, 298(16):385–388, 1984.
- [Hur89] Adolf Hurwitz. Über eine besondere Art der Kettenbruch-Entwicklung reeller größen. *Acta Math.*, 12:367–405, 1889.
- [IS06] Hiroyuki Inou and Mitsuhiro Shishikura. The renormalization for parabolic fixed points and their perturbation, 2006. Preprint.
- [Khi97] Aleksandr Y. Khinchin. *Continued fractions*. Dover Publications, Inc., Mineola, NY, 1997. With a preface by B. V. Gnedenko. Translated from the third (1961) Russian edition. Reprint of the 1964 translation.
- [Lan88] Oscar E. Lanford. *Renormalization Group Methods for Circle Mappings*, pages 25–36. Springer US, 1988.
- [Lim26a] Willie Rush Lim. Hyperbolicity of renormalization of critical quasicircle maps. *Commun. Math. Phys.*, 407(88), 2026.
- [Lim26b] Willie Rush Lim. Lebesgue measure of the postcritical set of neutral quadratic polynomials, 2026. Manuscript.
- [McM98] Curtis McMullen. Self-similarity of Siegel disks and Hausdorff dimension of Julia sets. *Acta Math.*, 180(2):247–292, 1998.
- [Min73] Bernhard Minnigerode. Über eine neue methode, die pellsche gleichung aufzulösen. *Nachr. Göttingen*, pages 619–653, 1873.
- [Nak81] Hitoshi Nakada. Metrical theory for a class of continued fraction transformations and their natural extension. *Tokyo J. Math.*, 4(2):399–426, 1981.
- [Pet96] Carsten Lunde Petersen. Local connectivity of some Julia sets containing a circle with an irrational rotation. *Acta Math.*, 177:163–224, 1996.
- [SY24] Mitsuhiro Shishikura and Fei Yang. The high type quadratic Siegel disks are Jordan domains. *J. Eur. Math. Soc.*, 27(11):4501–4562, 2024.
- [Yam99] Michael Yampolsky. Complex bounds for renormalization of critical circle maps. *Ergod. Theory Dyn. Syst.*, 19(1):227–257, 1999.
- [Yam01] Michael Yampolsky. The attractor of renormalization and rigidity of towers of critical circle maps. *Commun. Math. Phys.*, 218(3):537–568, 2001.
- [Yam02] Michael Yampolsky. Hyperbolicity of renormalization of critical circle maps. *Publ. Math. Inst. Hautes Études Sci.*, 96:1–41, 2002.
- [Yam03] Michael Yampolsky. Renormalization horseshoe for critical circle maps. *Commun. Math. Phys.*, 240(1–2):75–96, 2003.
- [Yoc95] Jean-Christophe Yoccoz. Théorème de Siegel, nombres de Bruno et polynômes quadratiques. In *Petits diviseurs en dimension 1*, number 231 in Astérisque, pages 1–88. Société mathématique de France, 1995.

DEPT. OF MATHEMATICS, BROWN UNIVERSITY, RI 02912

Email address: willie_rush_lim@brown.edu